Indefinite Integration

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Indefinite Integration

Differentials :

Up to this point in our work, for $y = f(x)$, we have regarded dy/dx as a composite symbol for the derivative $f'(x)$, whose component parts, dy and dx, had no meaning by themselves. It is now convenient to modify this point of view and attach meaning to dy and dx, so that thereafter we can treat dy/dx as though it were a fraction in fact as well as in appearance. We shall not however enter into any discussions on it. We shall only state that,

for a function of a single variable $y = f(x)$, the differential of y denoted by dy is the product of the derivative of y (with respect to x) and the differential of x denoted by dx. Thus,

Differential of $y = f(x)$ is $dy = f'(x)dx$.
For $y = x^4$, $dy = 4x^3\,dx$, or simply $d(x^4) = 4x^3\,dx$. Thus
$d(\sin x) = \cos x\,dx$, $d(y)^2 = 2y\,dy$, $d(\tan u) = \sec^2 u\,du$.

Integration as anti-derivative :

Simplest way to define integration is as an antiderivative or the inverse of a derivative of sin x is cos x then we may say that integral of cos x is sin x.

In general, if we consider

$$\frac{d}{dx} f(x) = \phi(x)$$

or, using differentials $d f(x) = \varphi(x)\,dx$:
then an integral of $\varphi(x)$ with respect to x or an integral of $\varphi(x)dx$ is $f(x)$ and symbolically, we write,

$$\boxed{\int \phi(x)\,dx = f(x)}$$

Where the symbol $\int$ which is an elongated S(the first letter of the word sum, or , of the latin word summa) is known as the sign of integration. Now we come to some formal definitions.

The actual process of finding the function is called **integration** where its derivative or its differential is known. The function to which the integration is applied is called **integrand** and the function obtained as a result of integration is said to be integral or **Anti-derivative or primitive**. In the above case, $\varphi(x)$ is the integrand and $f(x)$ is the integral.
The process of integrating many ordinary functions is simple, but in general, integration is more involved than differentiation, as will be evident from future discussions.

Some Basic Example

(i) $y = x^4$

$\Rightarrow \frac{dy}{dx} = 4x^3$

$\Rightarrow dy = 4x^3\,dx$

$\Rightarrow y = \int 4x^3 dx$

(ii) $y = x^2 + c$

$\Rightarrow \frac{d}{dx}\left(x^2 + c\right) = 2x$

$\Rightarrow \int 2x dx = x^2 + c$

(iii) $y = \sin x$

$\frac{dy}{dx} = \frac{d(\sin x)}{dx} = \cos x$

$\Rightarrow \cos x\,dx = d(\sin x)$

$\Rightarrow \int \cos x dx = \sin x + C$

(iv) $y = \tan x$

$\frac{dy}{dx} = \sec^2 x$

$\Rightarrow \int \sec^2 x dx = \tan x + C$

Know the facts



Integration can be said to be reverse of finding the $\frac{d}{dx}$.

Points to Remember!!!

If $\frac{d}{dx}[F(x)+C] = f(x)$ then $F(x) + C$ is called an antiderivative of $f'(x)$ on [a, b] and is written as $\int f(x)dx = F(x) + C$

Here C is called constant of Integration.

In this case we say that the function f(x) is integrable on [a, b]. Note that every function is not integrable

e.g. $f(x) = \begin{cases} 0 \text{ if } & x \in Q \\ 1 \text{ if } & x \in Q \end{cases}$ is not integrable in [0, 1]. Every function which is continuous on a closed and bounded interval is integrable.
However for integrability function f(x) may only be piece wise continuous in (a, b).

(v) $y = \ln x$

$\Rightarrow \frac{dy}{dx} = \frac{1}{x}$

$\Rightarrow \int \frac{1}{x} dx = \ln x + C$

(vi) $\int 4\tan^3 x \, d(\tan x) = (\tan x)^4 + C$

Here, integration is done with respect to 'tanx'.

Notes on Indefinite Integration :

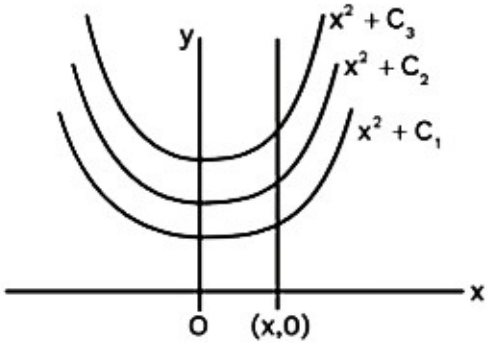
(1) Geometrical interpretation :

$y = \int 2x dx = x^2 + C$

$y = \int f(x) dx = F(x) + C$

$\Rightarrow F'(x) = f(x); F'(x_1) = f(x_1)$

Hence $y = \int f(x) dx$ denotes a **family of curves** such that the slope of the tangent at $x = x_1$ on every member is same. i.e. $F'(x_1) = f(x)$ (when x_1 lie in the domain of f(x)). Hence antiderivative of a function is not unique. If $g_1(x)$ and $g_2(x)$ are two antiderivatives of a function f(x) on [a, b] then they differ only by a constant i.e. $g_1(x) - g_2(x) = C$



(2) Antiderivative of a continuous function is differentiable:

i.e. if f(x) is continuous then $\int f(x) dx = F(x) + C \Rightarrow F'(x) - f(x) \Rightarrow F'(x)$ always exists

$\Rightarrow F(x)$ is differentiable.

(3) If integrand is discontinuous at $x = x_1$ then its antiderivative at $x = x_1$ need not be discontinuous.

i.e. e.g. $x^{-1/3}$ is discontinuous at x = 0.

But $\int x^{-1/3} dx = \frac{3}{2} x^{2/3} + C$ is continuous at x = 0

(4) If $\frac{d}{dx}(F(x)+C) = f(x) \Rightarrow \int f(x)dx = F(x) + C$

then only we say that f(x) is integrable.

(5) Antiderivative of a periodic function need not be a periodic function.

e.g. $f(x) = \cos x + 1$ is periodic but $\int(\cos x + 1)dx = \sin x + x + C$ is aperiodic.

Standard Integration

There are some functions which are frequently used in differentiation as well as integration. So, using these integrals (often called **'LOVING INTEGRAND'**), we can integrate much more complex functions.

1. (a) $\int x^n dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$.

(b) $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C,\ n \neq -1$

Proof: (a) $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = \frac{1}{(n+1)}(n+1)x^n = x^n$

Proof: (b)

$$\frac{d}{dx}\left(\frac{(ax+b)^{n+1}}{n+1}\right) = \left(\frac{1}{(n+1)}\right)(n+1)(ax+b)^n \times a$$

$= a\,(ax + b)^n$

So, we have to divide by 'a' to neutralize it.

Q. **Solve the following :**

(i) $\int e^{\ln\sqrt{x}}dx$

Sol. $\int e^{\ln\sqrt{x}}dx$

$= \int \sqrt{x}\,dx$

$= \int x^{1/2}dx$

$= \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$

Q. **(ii)** $\int e^{-\ln x^2}dx$

Sol. $\int e^{-\ln x^2}dx$

$= \int \frac{1}{x^2}dx = \int x^{-2}dx$

$= \frac{x^{-1}}{-1} + c = -\frac{1}{x} + c$

Q. **(iii)** $\int \ln\left(\frac{1}{e^x}\right)dx$

Sol. $\int \ln\left(e^{-x}\right)dx$

$= \int -x\,dx$

$= \frac{-x^2}{2} + c$

Q. **(iv)** $\int \frac{dx}{2\sqrt{x}}$

Sol. $\int \frac{dx}{2\sqrt{x}} = \frac{1}{2}\int x^{-1/2}dx$

$= \frac{1}{2}\left(\frac{x^{1/2}}{1/2}\right) + c$

$= \sqrt{x} + c$

Q. **(v)** $\int e^{\ln 2+\ln x}dx$

Sol. $\int e^{\ln 2x}dx$

$= \int 2x\ dx$

$= x^2 + c$

Q. **(vi)** $\int e^{m\ln x}dx$

Sol. $\int e^{\ln x^m}dx$

$= \int x^m\ dx$

$= \frac{x^{m+1}}{m+1} + C$

Q. **(vii)** $\int 2^{\ln x}dx$

Sol. $\int x^{\ln 2}\ dx \qquad \left(a^{\log_b c} = c^{\log_b a}\right)$

$= \frac{x^{(\ln 2+1)}}{\ln 2+1} + C$

2. **(a)** $\int \frac{1}{x}dx = \ln|x| + C$ (loving integrands)

(b) $\int \frac{dx}{(ax+b)} = \frac{\ln|ax+b|}{a} + C$

Proof :

(a) $\frac{d(\ln|x|)}{dx} = \frac{1}{|x|}\text{sgn}(x) = \frac{1}{x}$

(b) $\frac{d}{dx}\ln|ax+b| = \frac{a}{|ax+b|}\text{sgn}(ax+b) = \frac{a}{ax+b}$

Again, here we have 'a' in denominator to neutralize the 'a' which would come in numerator after differentiating ln (ax + b).

Q. **Solve the following :**

(i) $\int \frac{\sqrt{x^4 + x^{-4} + 2}}{x^3} dx$

Sol.

$$\int \frac{\sqrt{\left(x^2 + \frac{1}{x^2}\right)^2}}{x^3} dx$$

$$= \int \frac{x^2 + 1/x^2}{x^3} dx$$

$$= \int \frac{1}{x} + \frac{1}{x^5} dx$$

$$= \ln|x| + \frac{x^{-4}}{-4} + c \qquad (\because \int (f+g)dx = \int f dx + \int g dx)$$

Point to Remember!!!

Solving Trick:
Try to convert the given form into or very close to one of standard forms.

Q. **(ii)** $\int \frac{dx}{3 - 2x}$

Sol. $\int \frac{dx}{3-2x} = \frac{\ln|3-2x|}{-2} + c$

Q. **(iii)** $\frac{xdx}{a + bx}$

Sol.

$$\int \frac{xdx}{a+bx} = \frac{1}{b}\int \frac{bx\,dx}{a+bx}$$

$$= \frac{1}{b}\int \left(\frac{a+bx}{a+bx} - \frac{a}{a+bx}\right)dx$$

$$= \frac{1}{b}\int 1 - \frac{a}{a+bx} dx$$

$$= \frac{x}{b} - \left(\frac{a}{b}\right)\ln|a+bx| \times \left(\frac{1}{b}\right) + c$$

$$= \frac{x}{b} - \frac{a}{b^2}\ln|a+bx| + c$$

Q. **(iv)** $\int \frac{2x}{3x+5} dx$

Sol.

$$\int \frac{2x}{3x+5} dx = \frac{2}{3}\int \frac{3x+5-5}{3x+5} dx$$

$$= \frac{2}{3}\int 1 - \frac{5}{3x+5} dx$$

$$= \frac{2x}{3} - \frac{10}{3} \times \frac{\ln|3x+5|}{3} + c$$

$$= \frac{2x}{3} - \frac{10}{9} \ln|3x+5| + c$$

Q. **(v)** $\int \frac{x dx}{x^2 + 2x + 1}$

Sol. $\int \frac{x dx}{(x+1)^2}$

$$= \int \left[\frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right] dx$$

$$= \int \frac{1}{x+1} dx - \int \frac{dx}{(x+1)^2}$$

$$= \ln|x+1| + \frac{1}{x+1} + c$$

Q. **(vi)** $\int \frac{dx}{\sqrt{2x-3} + \sqrt{2x+3}}$

Sol. $\int \frac{\sqrt{2x-3} - \sqrt{2x+3}}{-6} dx$ (∵ rationalization)

$$= \int \frac{\sqrt{2x+3}}{6} dx - \int \frac{\sqrt{2x-3}}{6} dx$$

$$= \frac{(2x+3)^{3/2}}{6 \times \frac{3}{2} \times 2} - \frac{(2x-3)^{3/2}}{6 \times \frac{3}{2} \times 2} + c$$

$$= \frac{(2x+3)^{3/2} - (2x-3)^{3/2}}{18} + c$$

Q. **(vii)** $\int \frac{(1+x)^3}{x} dx$

Sol. $\int \frac{(1+x)^3}{x} dx$

$$= \int \frac{1 + 3x + 3x^2 + x^3}{x} dx$$

$$= \ln|x| + 3x + \frac{3x^2}{2} + \frac{x^3}{3} + c$$

Q. **(viii)** $\int \frac{1}{x(x+1)}dx$

Sol. $\int \frac{1}{x(x+1)}$ is of the form $\int \frac{1}{\text{quadratic}}dx$. In this case,

$\int \frac{dx}{x(x+1)}$ roots are 0 and –1

$$= \int \frac{(x+1)-(x+0)}{x(x+1)}dx$$

$$= \int \left(\frac{1}{x} - \frac{1}{x+1}\right)dx$$

$$= \ln|x| - \ln|x+1| + c$$

$$= \ln\left(\frac{|x|}{|x+1|}\right) + C$$

Point to Remember!!!

If quadratic has roots α and β then multiply and divide by the difference of α and β.

Q. **(ix)** $\int \frac{1}{(x+2)(x-3)}dx$

Sol. $\int \frac{1}{5}\frac{(x+2)-(x-3)}{(x+2)(x-3)}dx$

$$= \frac{1}{5}\int \left(\frac{1}{x-3} - \frac{1}{x+2}\right)dx$$

$$= \frac{1}{5}\ln\left|\frac{x-3}{x+2}\right| + c$$

Q. **(x)** $\int \frac{1}{(x+n)(x-n)}dx$ or $\int \frac{dx}{x^2-n^2}$

Sol. $\frac{1}{2n}\int \frac{(x+n)-(x-n)}{(x+n)(x-n)}dx$

$$= \frac{1}{2n}\ln\left|\frac{x-n}{x+n}\right| + c$$

Q. **(xi)** $\int \frac{x}{(x+2)(x+3)}\,dx$

Sol. $\int \frac{x}{(x+2)(x+3)}\,dx$

$$= \int \frac{3(x+2)-2(x+3)}{(x+2)\,(x+3)}\,dx$$

$$= \int \left(\frac{3}{x+3} - \frac{2}{x+2}\right)dx$$

$$= 3\ln|x+3| - 2\ln|x+2| + c$$

Note :

Because of how frequently we use it, the following integration is strongly recommended to remember as standard integration.

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c$$

Q. **Solve** $\int \frac{dx}{x^2-5x+6}dx$

Sol. **Method-I:**

$$\int \frac{(x-2)-(x-3)}{(x-2)(x-3)}dx$$

$$= \ln\left|\frac{x-3}{x-2}\right| + c$$

Method-II:

$$\int \frac{dx}{\left(x-\frac{5}{2}\right)^2 - \frac{1}{4}}$$

$$= \int \frac{dx}{\left(x-\frac{5}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \qquad \left(\text{form } \int \frac{dx}{x^2-a^2}, a = \frac{1}{2}\right)$$

$$= \frac{1}{2\times\left(\frac{1}{2}\right)} \ln\left|\frac{\left(x-\frac{5}{2}\right)-\frac{1}{2}}{\left(x-\frac{5}{2}\right)+\frac{1}{2}}\right| + c$$

$$= \ln\left|\frac{x-3}{x-2}\right| + c$$

3. (a) $\int e^x dx = e^x + c$

(b) $\int e^{(ax+b)} dx = \frac{e^{(ax+b)}}{a} + c$ (LOVING INTEGRANDS)

4. (a) $\int a^x dx = \frac{a^x}{\ln(a)} + c$

(b) $\int a^{(px+q)} dx = \frac{a^{(px+q)}}{p\ln a} + c,\ a > 0$

Proof : $\int a^{(px+q)}\, dx$

$$= \int e^{(px+q)\ln a} dx$$

$$= \frac{e^{(px+q)\ln a}}{p\ln a} + c = \frac{a^{px+q}}{p\ln a} + c$$

Q. **Solve the following :**

(i) $\int \frac{2^{x+1} - 5^{x+1}}{10^x} dx$

Sol. $\int \left(2.5^{-x} - 5.2^{-x}\right) dx$

$$= \left(\frac{2}{-\ln 5}\right)5^{-x} - \left(\frac{5}{-\ln 2}\right)2^{-x} + c$$

$$= \frac{5.2^{-x}}{\ln 2} - \frac{2}{\ln 5}.5^{-x} + c$$

Q. **(ii)** $\int a^x.e^x\, dx$

Sol. $\int (ae)^x dx = \frac{(ae)^x}{\ln(ae)} + c$

$$= \frac{(ae)^x}{1+\ln a} + c$$

Q. **(iii)** $\int \left(2^x + 3^x\right)^2 dx$

Sol. $\int \left(2^{2x} + 3^{2x} + 2.6^x\right) dx$

$$= \frac{4^x}{\ln 4} + \frac{9^x}{\ln 9} + \frac{2.6^x}{\ln 6} + c$$

Q. **(iv)** $\int \frac{e^{3x} + e^{5x}}{e^x + e^{-x}} dx$

Sol. $\int \frac{e^{4x}\left(e^x + e^{-x}\right)}{\left(e^x + e^{-x}\right)} dx$

$$= \int e^{4x} dx = \frac{e^{4x}}{4} + c$$

Q. **(v)** $\int a^{mx}.b^{nx}\ dx$

Sol. $\int \left(a^m.b^n\right)^x dx$

$$= \frac{\left(a^m.b^n\right)^x}{\ln\left(a^m.b^n\right)} + c$$

$$= \frac{a^{mx}b^{nx}}{m\ \ln a + n\ \ln b} + c$$

5. **(a)** $\int \sin x\ dx = -\cos x + c$

(b) $\int \sin\left(ax + b\right) dx = -\frac{\cos\left(ax + b\right)}{a} + c$

(c) $\int \cos x dx = \sin x + c$

(d) $\int \cos\left(ax + b\right) dx = \frac{\sin\left(ax + b\right)}{a} + c$

Q. **Solve the following :**

(i) $\int \sqrt{1 + \sin x}\ dx$

Sol. $\int \left|\sin\frac{x}{2} + \cos\frac{x}{2}\right| dx$

Case-I: $\sin\frac{x}{2}+\cos\frac{x}{2} \geq 0$

$$\Rightarrow \int\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)dx$$

$$=-2\cos\frac{x}{2}+2\sin\frac{x}{2}+c$$

Case-II: $\sin\frac{x}{2}+\cos\frac{x}{2}<0$

$$\Rightarrow \int-\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)dx$$

$$=2\cos\frac{x}{2}-2\sin\frac{x}{2}+c$$

Q. **(ii)** $\int \cos 2x \cos 3x\, dx$

Sol. $\int \cos 2x \cos 3x\, dx$

$$=\frac{1}{2}\int(\cos 5x+\cos x)\,dx$$

$$=\frac{1}{2}\left(\frac{\sin 5x}{5}+\sin x+c\right)$$

$$=\frac{\sin 5x}{10}+\frac{\sin x}{2}+c$$

Q. **(iii)** $\int\frac{\cos x-\cos 2x}{1-\cos x}dx$

Sol. $\int\frac{\cos x-\left(2\cos^2 x-1\right)}{1-\cos x}dx$

$$=\int\frac{-2\cos^2 x+\cos x+1}{1-\cos x}dx$$

$$=\int\frac{(1-\cos x)(2\cos x+1)}{1-\cos x}dx$$

$$=\int 2\cos x+1\,dx$$

$$=2\sin x+x+c$$

Q. **(iv)** $\int \sin^2 x \, dx$

Sol. $\int \left(\frac{1-\cos 2x}{2}\right) dx$

$= \frac{x}{2} - \frac{\sin 2x}{4} + c$

Q. **(v)** $\int \cos^2 x \, dx$

Sol. $\int \frac{1+\cos 2x}{2} \, dx$

$= \frac{x}{2} + \frac{\sin 2x}{4} + c$

Q. **(vi)** $\int \cos^3\theta \, d\theta$

Sol. $\int \frac{\cos 3\theta + 3\cos\theta}{4} d\theta$ $\quad (\because \cos 3\theta = 4\cos^3\theta - 3\cos\theta)$

$= \frac{1}{4}\left(\frac{\sin 3\theta}{3} + 3\sin\theta\right) + c$

Q. **(vii)** $\int \sin^4 x dx$

Sol. $\int \left(\frac{1-\cos 2x}{2}\right)^2 dx$

$= \frac{1}{4}\int \left(1 + \cos^2 2x - 2\cos 2x\right) dx$

$= \frac{1}{4}\int \left(1 + \left(\frac{1+\cos 4x}{2}\right) - 2\cos 2x\right) dx$

$= \frac{1}{4}\left(\frac{3x}{2} + \frac{\sin 4x}{8} - \sin 2x\right) + c$

$= \frac{3x}{8} + \frac{\sin 4x}{32} - \frac{\sin 2x}{4} + c$

Q. **(viii)** $\int \frac{\cos 5x + \cos 4x}{1 - 2\cos 3x}\, dx$

Sol.

$$\int \frac{2\cos\frac{9x}{2}\cos\frac{x}{2}}{1 - 2\left(2\cos^2\frac{3x}{2} - 1\right)}dx$$

$$= \int \frac{2\cos\left(\frac{9x}{2}\right)\cos\left(\frac{x}{2}\right)}{3 - 4\cos^2\frac{3x}{2}} \times \left(\frac{\cos\frac{3x}{2}}{\cos\frac{3x}{2}}\right)dx$$

$$= \int \frac{2\cos\left(\frac{9x}{2}\right)\cos\left(\frac{3x}{2}\right)\cos\left(\frac{x}{2}\right)}{-\cos\left(\frac{9x}{2}\right)}dx$$

$$= -\int (\cos x + \cos 2x)\, dx$$

$$= -\sin x - \frac{\sin 2x}{2} + c$$

Q. **(ix)** $\int \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x}dx$

Sol. $\int (\cos^2 x + \sin^2 x - \cos x \sin x)\, dx$

$$= \int \left(1 - \frac{\sin 2x}{2}\right)dx = x + \frac{\cos 2x}{4} + c$$

Q. **(x)** $\int \sin(x^\circ)\, dx$

Sol. $\int \sin\left(\frac{\pi x}{180}\right)dx \qquad \left(\because 1^\circ = \left(\frac{\pi}{180}\right)^c\right)$

$$= -\cos\left(\frac{\pi x}{180}\right) \times \frac{1}{\pi / 180} + c$$

$$= -\frac{180}{\pi}\cos(x^\circ) + c$$

6. **(a)** $\int \sec^2 x dx = \tan x + c$

(b) $\int \sec^2 (ax+b)\, dx = \frac{1}{a}\tan(ax+b) + c$

(c) $\int \operatorname{cosec}^2 x\, dx = -\cot x + c$

(d) $\int \operatorname{cosec}^2 (ax+b) dx = -\frac{1}{a}\cot (ax+b) + c$

Q. **Solve the following :**

(i) $\int \frac{dx}{1+\cos x}$

Sol. **Method-I:**

$$\int \frac{dx}{2\cos^2\left(\frac{x}{2}\right)} = \int \left(\frac{1}{2}\right)\sec^2\left(\frac{x}{2}\right)dx$$

$$= \left(\frac{1}{2}\right)\frac{\tan\left(\frac{x}{2}\right)}{\left(\frac{1}{2}\right)} + c$$

$$= \tan\left(\frac{x}{2}\right) + c$$

Method-II:

$$\int \frac{1-\cos x}{(1+\cos x)(1-\cos x)} dx$$

$$= \int \frac{1-\cos x}{\sin^2 x} dx$$

$$= \int (\operatorname{cosec}^2 x - \cot x \operatorname{cosec} x) dx$$

$$= -\cot x + \operatorname{cosec} x + c$$

Q. **(ii)** $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx$

Sol.

$$\int \frac{2\cos^2 x - 1 + 2\sin^2 x}{\cos^2 x} dx$$

$$= \int \frac{1}{\cos^2 x} dx = \int \sec^2 x\, dx$$

$$= \tan x + c$$

Q. **(iii)** $\int \frac{1-\cos x}{1+\cos x} dx$

Sol. $\int \frac{2\sin^2(x/2)}{2\cos^2(x/2)} dx$

$= \int \tan^2(x/2) dx$

$= \int \left(\sec^2(x/2) - 1\right) dx$

$= \frac{\tan(x/2)}{1/2} - x + c$

$= 2\tan(x/2) - x + c$

Q. **(iv)** $\int \cot^2 x \, dx$

Sol. $\int \cot^2 x \, dx$

$= -\cot x - x + c$

Q. **(v)** $\int \sec^2 x \operatorname{cosec}^2 x \, dx$

Sol. $\int \frac{1}{\sin^2 x \cos^2 x} dx$

$= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 \cos^2 x} dx$

$= \int \left(\sec^2 x + \operatorname{cosec}^2 x\right) dx$

$= \tan x - \cot x + c$

Q. **(vi)** $\int \cot^2 x \cos^2 x \, dx$

Sol. $\int \cot^2 x \left(1 - \sin^2 x\right) dx$

$= \int \left(\cot^2 x - \cos^2 x\right) dx$

$= \int \cot^2 x dx - \int \cos^2 x dx \quad = \frac{\sin 2x}{4} - \cot x - \frac{x}{2} + c$

Q. **(vii)** $\int \tan^2 x \, . \sin^2 x \, dx$

Sol. $\int \tan^2 x \left(1 - \cos^2 x\right) dx$

$= \int \left(\tan^2 x - \sin^2 x\right) dx \quad = \int \left(\sec^2 x - 1\right) - \left(\frac{1-\cos 2x}{2}\right) dx$

$= \tan x + \frac{\sin 2x}{4} - \frac{3x}{2} + c$

7. Now use can solve it.

(a) $\int \sec x \tan x \, dx = \sec x + c$

(b) $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

(c) $\int \sec(ax+b)\tan(ax+b)\,dx = \dfrac{\sec(ax+b)}{a} + c$

Q. **Solve the following :**

(i) $\int \dfrac{a\sin^3 x + b\cos^3 x}{\sin^2 \cos^2 x} dx$

Sol. $\int \dfrac{a\sin^3 x + b\cos^3 x}{\sin^2 \cos^2 x} dx$

$= \int (a\sec x \tan x + b \operatorname{cosec} x \cot x)\, dx$

$= a \sec x - b \operatorname{cosec} x + c$

Q. **(ii)** $\int \dfrac{\operatorname{cosec} x + \tan^2 x + \sin^2 x}{\sin x} dx$

Sol. $\int (\operatorname{cosec}^2 x + \sec x \tan x + \sin x)\, dx$

$= -\cot x + \sec x - \cos x + c$

Q. **(iii)** $\int \dfrac{dx}{1 - \sin 3x}$

Sol. **Method-I:** $\int \dfrac{1+\sin 3x}{\cos^2 3x} dx$

$= \int (\sec^2 3x + \sec 3x \tan 3x)\, dx$

$= \dfrac{\tan 3x}{3} + \dfrac{\sec 3x}{3} + c$

Method-II: $\int \dfrac{dx}{1+\cos\left(\frac{\pi}{2} + 3x\right)} = \dfrac{1}{2}\int \dfrac{dx}{\cos^2\left(\frac{3x + \frac{\pi}{2}}{2}\right)}$

$= \dfrac{1}{3}\tan\left(\dfrac{3x}{2} + \dfrac{\pi}{4}\right) + c$

Q. **(iv)** $\int \frac{\sin x}{\cos^2 x}\left(1-3\sin^3 x\right)dx$

Sol. $\int \tan x \sec x dx - 3\int \frac{\sin^4 x}{\cos^2 x}dx$

$$= \sec x - 3\int \frac{\left(1-\cos^2 x\right)^2}{\cos^2 x}dx$$

$$= \sec x - 3\int\left(\frac{1+\cos^4 x-2\cos^2 x}{\cos^2 x}\right)dx$$

$$= \sec x - 3\left(\tan x + \int \cos^2 x dx - 2x\right)$$

$$= \sec x - 3\tan x + 6x - 3\left(\frac{x}{2}+\frac{\sin 2x}{4}\right)+c$$

$$= \sec x - 3\tan x + \frac{9x}{2} - \frac{3\sin 2x}{4}+c$$

8. (a) $\int \frac{dx}{1+x^2} = \tan^{-1}x + c$

(b) $\int \frac{dx}{a^2+x^2} = \left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)+c$

9. (a) $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}x + c$

(b) $\int \frac{-dx}{\sqrt{1-x^2}} = \cos^{-1}x + c$

(c) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+c$

10. (a) $\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1}x + c$

(b) $\int \frac{dx}{|x|\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right)+c$

Proof :

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a^2}\int \frac{dx}{1+\left(\frac{x}{a}\right)^2}$$

$$= \frac{1}{a}.\int\left(\frac{1}{a}\right)\frac{1}{1+\left(\frac{x}{a}\right)^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$$

Q. **Solve the following :**

(i) $\int\left(\frac{x^2+\cos^2 x}{x^2+1}\right)\text{cosec}^2 x\,dx$

Sol. $\int\left(\frac{x^2+1-1+\cos^2 x}{x^2+1}\right)\text{cosec}^2 x\,dx$

$$= \int\left(\text{cosec}^2 x - \frac{1}{x^2+1}\right)dx$$

$$= -\cot x - \tan^{-1}x + c$$

Q. **(ii)** $\int\frac{x^2}{1+x^2}\,dx$

Sol. $\int\left(1-\frac{1}{1+x^2}\right)dx$

$$= x - \tan^{-1} x + c$$

Q. **(iii)** $\int\frac{x^4}{1+x^2}dx$

Sol. $\int\left(\frac{x^4-1}{1+x^2}+\frac{1}{1+x^2}\right)dx$

$$= \int\left(x^2-1+\frac{1}{1+x^2}\right)dx$$

$$= \frac{x^3}{3} - x + \tan^{-1}x + c$$

Q. **(iv)** $\int \frac{dx}{\sqrt{9-4x^2}}$

Sol. It is little bit similar to form $\int \frac{1}{\sqrt{a^2-x^2}}dx$.

So, $\frac{1}{2}\int \frac{dx}{\sqrt{\frac{9}{4}-x^2}}$, Now $a = \frac{3}{2}$

$$= \frac{1}{2}\sin^{-1}\left(\frac{x}{\left(\frac{3}{2}\right)}\right)+c = \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right)+c$$

Q. **(v)** $\int \frac{dx}{x^2-x+1}dx$

Sol. $$\int \frac{dx}{\left(x-\frac{1}{2}\right)^2+\frac{3}{4}}$$

It is of the form $\int \frac{dx}{x^2+a^2}, x \to x-\frac{1}{2}, a \to \frac{\sqrt{3}}{2}$

$$\Rightarrow \frac{1}{\left(\frac{\sqrt{3}}{2}\right)}\tan^{-1}\left(\frac{\left(x-\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)}\right)+c$$

$$= \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)+c$$

Q. **(vi)** $\int \frac{dx}{49+25x^2}dx$

Sol. $$\int\left(\frac{1}{25}\right)\frac{dx}{\left(\left(\frac{7}{5}\right)^2+x^2\right)}$$

$$= \frac{1}{25}\times\frac{1}{(7/5)}\times\tan^{-1}\frac{x}{(7/5)}+c$$

$$= \frac{1}{35}\tan^{-1}\left(\frac{5x}{7}\right)+c$$

Q. (vii) $\int \frac{dx}{(2x-7)\sqrt{(x-3)(x-4)}}$

Sol. $\int \frac{2dx}{(2x-7)\sqrt{4x^2-28x+48}}$

$= \int \frac{2\,dx}{(2x-7)\sqrt{(2x-7)^2-1}}$

$= 2 \times \frac{\sec^{-1}(2x-7)+c}{2}$

$= \sec^{-1}(2x-7)+c$

Q. (viii) $\int \frac{dx}{(x^2-4x+4)(x^2-4x+5)}$

Sol. $\int \frac{(x^2-4x+5)-(x^2-4x+4)}{(x^2-4x+4)(x^2-4x+5)}dx$

$= \int \frac{dx}{x^2-4x+4} - \int \frac{dx}{x^2-4x+5}$

$= \int \frac{dx}{(x-2)^2} - \int \frac{dx}{(x-2)^2+1}$

$= -\left(\frac{1}{x-2}\right) - \tan^{-1}(x-2)+c$

$= \frac{1}{2-x} + \tan^{-1}(2-x)+c$

The Ultimate Shortcut

$$\int \frac{dx}{\text{Quadratic}} = \begin{cases} \tan^{-1}(\ ) & \text{if } D<0 \\ \ln(\ ) & \text{if } D>0 \\ \text{Rational function} & \text{if } D=0 \end{cases}$$

In case of $D > 0$, multiply and divide by difference of roots.

In case of $D < 0$, transform into sum of perfect square and constant.

Ex. $\int 2^x . e^x dx$

Sol. $\int 2^x e^x dx = \int (2e)^x dx = \frac{(2e)^x}{\ln(2e)} + c$

Ex. $\int \frac{1-\tan^2 x}{1+\tan^2 x} dx$

Sol. $\int \frac{1-\tan^2 x}{1+\tan^2 x} dx = \frac{1-\tan^2 x}{\sec^2 x} dx = \int (\cos^2 x - \sin^2 x) dx = \int \cos 2x dx = \frac{1}{2}\sin 2x + c$

Ex. $\int \frac{\left(\sqrt{x+1}\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} dx$

Sol. $\int \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} = \int \frac{\left(\sqrt{x}+1\right)\sqrt{x}\left(\left(\sqrt{x}\right)^3-1\right)}{\sqrt{x}\left(x+\sqrt{x}+1\right)} dx$

$= \int \frac{\left(\sqrt{x}+1\right)\left(\sqrt{x}-1\right)\left(x+1+\sqrt{x}\right)}{\left(x+\sqrt{x}+1\right)} dx = \int (x-1) dx = \frac{x^2}{2} - x + c$

Ex. $\int \frac{\left(x^2+\sin^2 x\right)\sec^2 x}{1+x^2} dx$

Sol. $\int \frac{\left(x^2+\sin^2 x\right)\sec^2 x}{1+x^2} dx = \int \left(\sec^2 x - \frac{1}{1+x^2}\right) dx = \tan x - \tan^{-1} x + c$

Ex. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$

Sol. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$

$= \frac{(\sin x + \cos x + \sin k + \cos k)(\sin x + \cos x - \sin k - \cos k)}{\sin x - \sin k + \cos x - \cos k} dx$

$= \int (\sin x + \cos x + \sin k + \cos k) dx = (\sin x - \cos x) + (\sin x + \cos k) x + C$

Q1 **Find f(x) if $f'(x^2) = \frac{1}{x}$ for x > 0, f(1) = 1.**

Sol.

$$\frac{d}{dx}f(x^2) = f'(x^2) \times 2x$$

$$\frac{d}{dx}f(x^2) = 2$$

$\Rightarrow f(x^2) = \int 2dx$

$\Rightarrow f(x^2) = 2x + c$

$\Rightarrow f(x) = 2\sqrt{x} + c$

Since $f(1) = 1 \Rightarrow c = -1$

$\therefore f(x) = 2\sqrt{x} - 1$

Q2 **Find f(x) if $f'(\sin^2 x) = \cos^2 x \ \forall \ x \in R$, f(1) = 1.**

Sol.

$f'(t) = 1 - t$

$\Rightarrow f(t) = \int (1-t)dt$

$$\Rightarrow f(t) = t - \frac{t^2}{2} + c$$

$$\Rightarrow 1 = 1 - \frac{1}{2} + c \quad (\because f(1) = 1)$$

$$\Rightarrow c = \frac{1}{2} \Rightarrow f(t) = t - \frac{t^2}{2} + \frac{1}{2}$$

$$\Rightarrow f(x) = x - \frac{x^2}{2} + \frac{1}{2}$$

Q3 **Find f(x) if $f'(\sin x) = \cos^2 x$ for all x, f(1) = 1**

Sol.

$f'(t) = 1 - t^2$

$\Rightarrow f(t) = \int (1-t^2)dt$

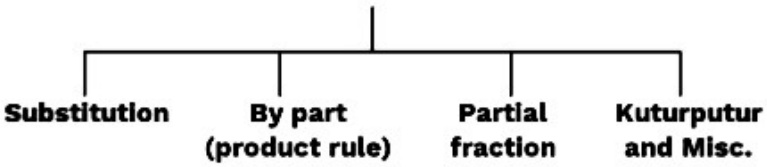
$$\Rightarrow f(x) = x - \frac{x^3}{3} + c$$

$$\Rightarrow c = \frac{1}{3} \ (\because f(1) = 1)$$

$$\Rightarrow f(x) = x - \frac{x^3}{3} + \frac{1}{3}$$

Techniques of integration :

Often it is not possible to convert an integral into loving integral just by simple manipulation. Then it requires some techniques to convert an integral into loving integral. These techniques are following.



Substitution:

Theory: $I = \int f(x)dx$ and let $x = \phi(z)$

$$\frac{dI}{dx} = f(x); \frac{dx}{dz} = \phi'(z); dx = \phi'(z)dz$$

$$\Rightarrow \frac{dI}{dz} = \frac{dI}{dx}.\frac{dx}{dz} = f(x).\phi'(z) \text{ or } \frac{dI}{dz} = f(\phi(z))\phi'(z)$$

Hence $I = \int f(\phi(z))\phi'(z)dz$ ...(1)

Substitution is said to be appropriate if the integrand in (1) is a loving one. (Standard integral)

Ex. $\int (f(x))^n f'(x)dx$

Sol. Let $f(x) = t$

$$\Rightarrow f'(x) = \frac{dt}{dx} \Rightarrow f'(x)\,dx = dt$$

$$\Rightarrow \int t^n dt$$

$$= \frac{t^{n+1}}{n+1} + c$$

$$= \frac{(f(x))^{n+1}}{n+1}$$

Ex. $\int \frac{f'(x)}{(f(x))^n} dx$

Sol. Again, Let $f(x) = t \Rightarrow f'(x)dx = dt$

$$\Rightarrow \int \frac{dt}{t^n} = \frac{t^{-n+1}}{-n+1} + c$$

$$\Rightarrow \frac{(f(x))^{-n+1}}{-n+1} + c$$

Q. **Solve the following :**

(i) $\int \tan x \sec^2 x \, dx$

Sol. Let $\tan x = t$ $\left(\because \frac{d(\tan x)}{dx} = \sec^2 x \text{ which is present} \right)$

$$\Rightarrow \sec^2 x = \frac{dt}{dx}$$

$$\Rightarrow \int t \, dt$$

$$= \frac{t^2}{2} + c$$

$$= \frac{\tan^2 x}{2} + c = \frac{\sec^2 x}{2} + c_1$$

Q. **(ii)** $\int \sqrt{\tan x} \sec^2 x \, dx$

Sol. Let $\tan x = t$

$$\Rightarrow \int \sqrt{t} \, dt = \frac{2t^{3/2}}{3} + c$$

$$= \frac{2}{3}(\tan x)^{3/2} + c$$

Q. **(iii)** $\int \sqrt{5 + \tan x} \sec^2 x \, dx$

Sol. Let $\tan x = t \Rightarrow \sec^2 x \, dx = dt$

$$\Rightarrow \int \sqrt{5 + t} \, dt$$

$$= \frac{2}{3}(5 + t)^{3/2} + c$$

$$= \frac{2}{3}(5 + \tan x)^{3/2} + c$$

Q. **(iv)** $\int \frac{\sqrt{\tan x}}{\sin 2x} dx$

Sol. $\int \frac{\sqrt{\tan x}}{2.\tan x.\cos^2 x} dx$

$= \int \frac{\sec^2 x}{2\sqrt{\tan x}} dx$

Let $\tan x = t$

$\therefore \int \frac{dt}{2\sqrt{t}} = \sqrt{t} + c = \sqrt{\tan x} + c$

Q. **(v)** $\int (\tan x - x)(\sec^2 x - 1) dx$

Sol. **Trick:** Differentiate parts of function to check for substitution.

We can see that $\frac{d(\tan x - x)}{dx} = \sec^2 x - 1$

$\Rightarrow (\sec^2 x - 1)\, dx = d(\tan x - x)$

So, lets put $\tan x - x = t$

$\Rightarrow \int t\, dt = \frac{t^2}{2} + c$

$= \frac{(\tan x - x)^2}{2} + c$

Q. **(vi)** $\int \tan^3 x - x\tan^2 x\, dx$

Sol. $\int \tan^2 x (\tan x - x)\, dx$

$\tan x - x = t$

$\Rightarrow (\sec^2 x - 1)\, dx = dt$

$\Rightarrow \tan^2 x\, dx = dt$

$\Rightarrow \int t dt = \frac{t^2}{2} + c$

$= \frac{(\tan x - x)^2}{2} + c$

Q. **(vii)** $\int \frac{\sec^4 x}{\sqrt{\tan x}} dx$

Sol. $\tan x = t \Rightarrow \sec^2 x\, dx = dt$

$$\Rightarrow \int \frac{1+t^2}{\sqrt{t}}dt$$

$$= \int\left(\frac{1}{\sqrt{t}} + t^{3/2}\right)dt$$

$$= 2\sqrt{t} + \frac{2}{5}t^{5/2} + c$$

$$= 2\sqrt{\tan x} + \frac{2}{5}(\tan x)^{5/2} + c$$

Q. **(viii)** $\int \frac{\tan\sqrt{x}\sec^2\sqrt{x}}{2\sqrt{x}}dx$

Sol. Let $\tan\sqrt{x} = t$

$$\Rightarrow \int t\,dt = \frac{t^2}{2} + c$$

$$= \frac{\tan^2\sqrt{x}}{2} + c$$

Q. **(ix)** $\int \frac{2x}{\sqrt{1-x^2}}dx$

Sol. Let $1 - x^2 = t$

$$\Rightarrow -2xdx = dt$$

$$\Rightarrow \int -\frac{dt}{\sqrt{t}} = -2\sqrt{t} + c$$

$$= -2\sqrt{1-x^2} + c$$

Q. **(x)** $\int \frac{2x - \sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}dx$

Sol. $\int \frac{2x}{\sqrt{1-x^2}}dx - \int \frac{\sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}dx$

$$= -2\sqrt{1-x^2} - \int \sqrt{\sin^{-1}x}\,d\left(\sin^{-1}x\right)\left(\because \frac{d\left(\sin^{-1}x\right)}{dx} = \frac{1}{\sqrt{1-x^2}}\right)$$

$$= -2\sqrt{1-x^2} - \frac{2}{3}\left(\sin^{-1}x\right)^{3/2} + c$$

Q. **(xi)** $\int \frac{e^{\sqrt{x}}\cos\left(e^{\sqrt{x}}\right)}{\sqrt{x}}dx$

Sol. Let $e^{\sqrt{x}} = t$

$\Rightarrow \frac{e^{\sqrt{x}}}{2\sqrt{x}}dx = dt$

$\Rightarrow \int 2\cos t\ dt = 2\sin t + c$

$= 2\sin\left(e^{\sqrt{x}}\right) + c$

Q. **(xii)** $\int e^{x}\sin\left(e^{x}\right)dx$

Sol. $e^{x} = t$

$\Rightarrow \int \sin t\ dt = -\cos t + c$

$= -\cos(e^{x}) + c$

Q. **(xiii)** $\int \frac{x^{7}}{1+x^{16}}dx$

Sol. Let $x^{8} = t$

$\Rightarrow 8x^{7}\,dx = dt$

$\Rightarrow \int \frac{1}{8}\left(\frac{1}{1+t^{2}}\right)dt$

$= \frac{\tan^{-1}t}{8} + c$

$= \frac{\tan^{-1}\left(x^{8}\right)}{8} + c$

Q. **(xiv)** $\int \frac{x^{2}\tan^{-1}\left(x^{3}\right)}{1+x^{6}}dx$

Sol. Let $\tan^{-1}(x^{3}) = t$

$\Rightarrow \frac{3x^{2}}{1+x^{6}}dx = dt$

$\Rightarrow \int \frac{t}{3}dt = \frac{t^{2}}{6} + c$

$= \frac{\left(\tan^{-1}\left(x^{3}\right)\right)^{2}}{6} + c$

Q. **(xv)** $\int \ln^2(\sec x)\tan x\,dx$

Sol. $\ln(\sec x) = t$

$$\Rightarrow \left(\frac{1}{\sec x}\right)(\sec x\tan x)\,dx = dt$$

$$\Rightarrow \int t^2 dt = \frac{t^3}{3} + c$$

$$= \frac{\ln^3|\sec x|}{3} + c$$

Q. **(xvi)** $\int \frac{\sin 2x}{1+2\sin^2 x}dx$

Sol. Let $\sin^2 x = t$

$\Rightarrow$ 2 sinx cos x dx = sin 2x dx = dt

$$\Rightarrow \int \frac{dt}{1+2t} = \frac{\ln|1+2t|}{2} + c = \frac{1}{2}\ln|1+2\sin^2 x| + c$$

Q. **(xvii)** $\int \frac{\sin 2x dx}{1+3\cos^2 x}$

Sol. Let $\cos^2 x = t$

$\Rightarrow$ $-\sin 2x\,dx = dt$

$$\Rightarrow \int \frac{-dt}{1+3t} = \frac{-\ln(1+3t)}{3} + c$$

$$= \frac{-\ln(1+3\cos^2 x)}{3} + c$$

Q. **(xviii)** $\int \frac{\sin 2x\,dx}{a\sin^2 x + b\cos^2 x}$

Sol. Let $\sin^2 x = t$

$$\Rightarrow \int \frac{dt}{at + b(1-t)}$$

$$= \int \frac{dt}{(a-b)t + b}$$

$$= \frac{\ln((a-b)t+b)}{(a-b)} + c$$

$$= \frac{\ln\left((a-b)\sin^2 x + b\right)}{a-b} + c$$

$$= \frac{\ln\left|a\sin^2 x + b\cos^2 x\right|}{a-b} + c$$

Q. **(xix)** $\int \frac{\tan x\, dx}{a + b\tan^2 x}$

Sol. $\int \frac{\sin x \cos x dx}{a\cos^2 x + b\sin^2 x}$

$$= \frac{\ln\left|a\cos^2 x + b\sin^2 x\right|}{2(b-a)} + c$$

Q. **(xx)** $\int \frac{1}{x^{1/2} + x^{1/3}} dx$

Sol. We generally prefer natural exponent on x for the case in integration.
Let's take L.C.M of 2 and 3 = 6

$\Rightarrow$ Let $x^{1/6} = y$

$\Rightarrow$ $y^6 = x$

$\Rightarrow$ $6y^5 dy = dx$

$\Rightarrow$ $\int \frac{6y^5 dy}{y^3 + y^2}$

$$= 6\int \frac{y^3}{y+1} dy$$

$$= 6\int \left(\frac{y^3+1}{y+1} - \frac{1}{y+1}\right) dy$$

$$= 6\left[\int (y^2 - y + 1) dy - \int \frac{dy}{y+1}\right]$$

$$= 6\left[\frac{y^3}{3} - \frac{y^2}{2} + y - \ln|y+1| + c\right]$$

$$= 2x^{1/2} - 3x^{1/3} + 6x^{1/6} - 6\ln\left|1 + x^{1/6}\right| + c$$

Q. **(xxi)** $\int \frac{dx}{x^{1/2} + x^{1/4}}$

Sol. $x^{1/4} = y$

$$\Rightarrow \int \frac{4y^3 dy}{y^2 + y}$$

$$= 4\int \frac{y^2}{y+1} dy$$

$$= 4\int \left(\frac{y^2 - 1}{y+1} + \frac{1}{y+1} \right) dy$$

$$= 4\left(\frac{y^2}{2} - y + \ln|y+1| + c \right)$$

$$= 2\sqrt{x} - 4x^{1/4} + 4\ln\left|1 + x^{1/4}\right| + c$$

Q. **(xxii)** $\int \frac{2x^5 - 3x^2}{1 + 3x^3 - x^6} dx$

Sol. Let $x^3 = t$

$$\Rightarrow \frac{1}{3}\int \frac{2t - 3}{1 + 3t - t^2} dt$$

Let $3t - t^2 = y$

$$\Rightarrow (3 - 2t)dt = dy$$

$$\Rightarrow -\frac{1}{3}\int \frac{dy}{1+y}$$

$$= -\frac{\ln|1+y|}{3} + c$$

$$= -\frac{\ln|1 + 3t - t^2|}{3} + c$$

$$= -\frac{\ln|1 + 3x^3 - x^6|}{3} + c$$

Q. **(xxiii)** $\int \left(1 - 3x^{4/3}\right)^{1/10} x^{1/3} dx$

Sol. $1 - 3x^{4/3} = t$

$$\Rightarrow -4x^{1/3} dx = dt$$

$$\Rightarrow -\frac{1}{4}\int t^{1/10} dt$$

$$= \left(-\frac{1}{4}\right)\frac{t^{\left(\frac{11}{10}\right)}}{\left(\frac{11}{10}\right)} + c$$

$$= \frac{-5t^{(11/10)}}{22} + c$$

$$= \frac{(-5)\left(1-3x^{4/3}\right)^{11/10}}{22} + c$$

Standard Integration

11. (a) $\int \tan x\,dx = \ln|\sec x| + c = -\ln|\cos x| + c$ (LOVING INTEGRAND)

(b) $\int \cot x\,dx = \ln|\sin x| + c$

Proof:

$$\int \frac{\sin x}{\cos x}\,dx$$

$\cos x = t$

$$\Rightarrow \int \frac{-dt}{t} = -\ln|t| + c$$

$$= -\ln|\cos x| + c = \ln|\sec x| + c$$

ILLUSTRATION :

Ex. $\int \frac{\cos(\ln x)}{x}\,dx$

Sol. Put $\ln x = t \Rightarrow \frac{1}{x}dx = dt$

Integral become $\int \cos t = \sin t + c = \sin(\ln x) + c$

Ex. $\int \frac{x^3 dx}{1+x^8}$

Sol. $x^4 = t \Rightarrow 4x^3 dx = dt$

Integral becomes $\int \frac{\frac{1}{4}dt}{1+t^2} = \frac{1}{4}\tan^{-1} t + c = \frac{1}{4}\tan^{-1}\left(x^4\right) + c$

Ex. $\int \frac{\ln\left(x+\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}dx$

Sol. Put $\ln\left(x+\sqrt{1+x^2}\right)=t \Rightarrow \frac{1+\frac{2x}{2\sqrt{1+x^2}}}{x+\sqrt{1+x^2}}dx=dt$ or $\frac{1}{\sqrt{1+x^2}}dx=dt$

Integral become $\int t\,dt=\frac{t^2}{2}+c=\frac{1}{2}\left(\ln\left\{x+\sqrt{1+x^2}\right\}\right)^2+c$

Ex. $\int \frac{\tan\sqrt{x}\sec^2\sqrt{x}}{\sqrt{x}}dx$

Sol. Put $\tan\sqrt{x}=t \Rightarrow \frac{\sec^2\sqrt{x}}{2\sqrt{x}}dx=dt$

Integral becomes $\int 2t\,dt=t^2+c=\left(\tan\sqrt{x}\right)^2+c$

Q. **Solve the following :**

(i) $\int \frac{\tan(\ln x)}{x}dx$

Sol. $\ln x = t$

$= \int \tan t\,dt = \ln|\sec t|+c$

$= \ln|\sec(\ln x)| + c$

Q. **(ii)** $\int \frac{\tan\left(\sin^{-1}x\right)}{\sqrt{1-x^2}}dx$

Sol. Let $\sin^{-1}(x) = t$

$\Rightarrow \int \tan t\,dt = \ln|\sec(\sin^{-1}x)| + c$

Q. **(iii)** $\int \frac{\cos x}{\cos(x-a)}dx$

Sol. $\int \frac{\cos((x-a)+a)}{\cos(x-a)}dx$

$= \int \frac{\cos(x-a)\cos a - \sin(x-a)\sin a}{\cos(x-a)}dx$

$= \int (\cos a - \tan(x-a)\sin a)dx$

$= x\cos a - \sin a \ln|\sec(x-a)| + c$

Q. **(iv)** $\int \frac{x\cos x}{(x\sin x + \cos x)^2}dx$

Sol. Let $x \sin x + \cos x = t$

$\Rightarrow$ $(x\cos x + \sin x - \sin x)\,dx = dt$

$\Rightarrow$ $\int \frac{dt}{t^2} = -\frac{1}{t} + c$

$= \frac{-1}{x\sin x + \cos x} + c$

Q. **(v)** $\int \frac{\sin 2x\,dx}{(\sin 5x).(\sin 3x)}$

Sol. $\int \frac{\sin(5x-3x)}{\sin 5x\ \sin 3x}dx$

$= \int (\cot 3x - \cot 5x)\,dx$

$= \frac{\ln|\sin 3x|}{3} - \frac{\ln|\sin 5x|}{5} + c$

Q. **(vi)** $\int (\sin x - \cos x)(\sin x + \cos x)^5\,dx$

Sol. Let $\sin x + \cos x = t$

$\Rightarrow$ $\cos x - \sin x = dt$

$\Rightarrow$ $\int -t^5 dt$

$= -\frac{t^6}{6} + c$

$= \frac{-(\sin x + \cos x)^6}{6} + c$

Q. **(vii)** $\int x\sqrt{\dfrac{2\sin\left(x^2+1\right)-\sin\left(2\left(x^2+1\right)\right)}{2\sin\left(x^2+1\right)+\sin\left(2\left(x^2+1\right)\right)}}\,dx$

Sol. $\int x\sqrt{\dfrac{1-\cos\left(x^2+1\right)}{1+\cos\left(x^2+1\right)}}\,dx$

$= \int x\tan\left(\dfrac{x^2+1}{2}\right)dx$

Let $\dfrac{x^2+1}{2} = y$

$\Rightarrow \int \tan y\,dy = \ln|\sec y| + c$

$= \ln\left|\sec\left(\dfrac{x^2+1}{2}\right)\right| + c$

Q. **(viii)** $\int \dfrac{x + e^x(\sin x + \cos x) + \sin x\cos x}{\left(x^2 + 2e^x\sin x - \cos^2 x\right)^2}\,dx$

Sol. Let $x^2 + 2e^x \sin x - \cos^2 x = t$

$\Rightarrow \dfrac{1}{2}\int \dfrac{dt}{t^2} = \dfrac{-1}{2t} + c$

$= \dfrac{-1}{2\left(x^2 + 2e^x\sin x - \cos^2 x\right)} + c$

12. (a) $\int \sec x\,dx = \ln(\sec x + \tan x) + c$ or $\ln\tan\left(\dfrac{\pi}{4} + \dfrac{x}{2}\right) + c$

(b) $\int \operatorname{cosec} x\,dx = \ln(\operatorname{cosec} x - \cot x) + c$ or $\ln\tan\dfrac{x}{2} + c$ } (LOVING INTEGRANDS)

Proof :

$\int \sec x\,dx = \int \dfrac{\sec x(\sec x + \tan x)}{\sec x + \tan x}\,dx$

Put $\sec x + \tan x = t$

$\Rightarrow$ $(\sec x \tan x + \sec^2 x)\,dx = dt$

Integral becomes $\int \dfrac{dt}{t} = \ln t = \ln(\sec x + \tan x) + c$

Ex. $\int \frac{\text{cosec}\left(\tan^{-1}x\right)}{1+x^2}dx$

Sol. Put $\tan^{-1} x = t \Rightarrow \frac{1\,dx}{1+x^2} = dt$

Integral becomes $\int \text{cosec}\, t\ dt = \ln\left(\text{cosec}\, t - \cot t\right) + c$

$= \ln[\text{cosec}(\tan^{-1}x) - \cot(\tan^{-1}x)] + c = \ln\left[\frac{\sqrt{1+x^2}}{x} - \frac{1}{x}\right] + c$

Ex. $\int \frac{\cos 2x}{\sin x}dx$

Sol. $\int \frac{1-2\sin^2 x}{\sin x}dx = \int \left(\text{cosec}\, x - 2\sin x\right)\ dx = \ln\left(\text{cosec}\, x - \cot x\right) + 2\cos x + c$

Ex. $\int \frac{e^x\left(1+x\right)}{\cos\left(xe^x\right)}dx$

Sol. Put $xe^x = t \Rightarrow e^x\,(1+x)\,dx = dt$

Integral becomes

$\int \frac{dt}{\cos t} = \int \sec t\, dt = \ln\left(\sec t + \tan t\right) + c = \ln\left(\sec\left(xe^x\right) + \tan\left(xe^x\right)\right) + c$

Q. **Solve the following :**

(i) $\int \frac{\sin x}{\sin 2x}dx$

Sol. $\int \frac{1}{2\cos x}dx = \int \frac{\sec x}{2}dx$

$= \frac{\ln\left|\sec x + \tan x\right|}{2} + c$

Q. **(ii)** $\int \frac{dx}{4\cos^3 x - 3\cos x}$

Sol. $\int \frac{dx}{\cos 3x} = \frac{\ln\left|\sec 3x + \tan 3x\right|}{3} + c$

Q. (iii) $\int \frac{\tan x + \sec x - 1}{\tan x - \sec x + 1} dx$

Sol. $\int \frac{\tan x + \sec x - \left(\sec^2 x - \tan^2 x\right)}{\tan x - \sec x + 1} dx$

$= \int \left(\tan x + \sec x\right) dx$

$= \ln|\sec x + \tan x| + \ln|\sec x| + c$

Q. (iv) $\int \frac{dx}{\sin x \cos^2 x}$

Sol. $\int \frac{\sin^2 x + \cos^2 x}{\sin x \cos^2 x} dx$

$= \int \left(\sec x \tan x + \operatorname{cosec} x\right) dx$

$= \sec x + \ln|\operatorname{cosec} x - \cot x| + c$

Q. (v) $\int \frac{dx}{\sqrt{3}\sin x + \cos x}$

Sol. $\frac{1}{2}\int \frac{dx}{\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x}$

$= \frac{1}{2}\int \frac{dx}{\cos\left(x - \frac{\pi}{3}\right)}$

$= \frac{1}{2}\ln\left|\sec\left(x - \frac{\pi}{3}\right) + \tan\left(x - \frac{\pi}{3}\right)\right| + c$

Note :

$$\int \frac{dx}{a\sin x + b\cos x} = \frac{1}{\sqrt{a^2 + b^2}} \int \frac{dx}{\frac{a}{\sqrt{a^2 + b^2}}\sin x + \frac{b}{\sqrt{a^2 + b^2}}\cos x}$$

$$= \frac{1}{\sqrt{a^2 + b^2}} \int \sec\left(x - \alpha\right) dx$$

$$= \frac{1}{\sqrt{a^2 + b^2}} \times \ln\left|\sec\left(x - \alpha\right) + \tan\left(x - \alpha\right)\right| + c$$

where $\alpha = \cos^{-1}\left(\frac{b}{\sqrt{a^2 + b^2}}\right)$

Q. **(vi)** $\int \frac{dx}{\sec x + \operatorname{cosec} x}$

Sol.

$$\int \frac{dx}{\frac{1}{\cos x} + \frac{1}{\sin x}}$$

$$= \int \frac{\sin x \cos x}{\sin x + \cos x} dx$$

$$= \frac{1}{2}\int \frac{(\sin x + \cos x)^2 - 1}{\sin x + \cos x} dx$$

$$= \frac{1}{2}\int (\sin x + \cos x) dx - \frac{1}{2}\int \frac{dx}{\sin x + \cos x}$$

$$= \frac{\sin x - \cos x}{2} - \frac{1}{2\sqrt{2}}\int \sec\left(x - \frac{\pi}{4}\right) dx$$

$$= \frac{\sin x - \cos x}{2} - \frac{\ln\left|\sec\left(x - \frac{\pi}{4}\right) + \tan\left(x - \frac{\pi}{4}\right)\right|}{2\sqrt{2}} + c$$

Q. **(vii)** $\int \frac{dx}{\tan x + \cot x + \sec x + \operatorname{cosec} x}$

Sol.

$$\int \frac{dx}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} + \frac{1}{\cos x} + \frac{1}{\sin x}}$$

$$= \int \frac{\sin x \cos x \, dx}{1 + \sin x + \cos x}$$

$$= \frac{1}{2}\int \frac{(\sin x + \cos x)^2 - 1}{(\sin x + \cos x) + 1} dx \qquad (\because (\sin x + \cos x)^2 = 1 + 2 \sin x \cos x)$$

$$= \frac{1}{2}\int (\sin x + \cos x - 1) dx$$

$$= \frac{-\cos x + \sin x - x}{2} + c$$

Q. **(viii)** $\int e^{(x^2 + \ln x)} dx$

Sol. $\int e^{x^2} . x \, dx$

let $x^2 = t$

$\Rightarrow \int \frac{e^t}{2} dt = \frac{e^t}{2} + c = \frac{e^{x^2}}{2} + c$

Q. **(ix)** $\int \frac{dx}{e^x + 1}$

Sol. $e^x = t$

Method-I: $\int \frac{dt}{t(t+1)}$

$$\int \left(\frac{1}{t} - \frac{1}{t+1}\right) dt$$

$$= \ln\left|\frac{t}{t+1}\right| + c$$

Method-II: $\int \frac{e^{-x}}{1+e^{-x}} dx$

Let $1 + e^{-x} = t$

$$\int -\frac{dt}{t} = -\ln|t| + c$$

$$= -\ln(1+e^{-x}) + c$$

Q. **(x)** $\int \frac{e^x - 1}{e^x + 1} dx$

Sol. $\int \frac{e^x}{e^x + 1} dx - \int \frac{1}{e^x + 1} dx$

$= \ln(e^x+1) + \ln(e^{-x} + 1) + c$

Q. **(xi)** $\int \frac{e^x(1+x)}{\sin^2(xe^x)} dx$

Sol. Let $x\, e^x = t$

$\Rightarrow$ $e^x (x + 1)\, dx = dt$

$\Rightarrow$ $\int \frac{dt}{\sin^2 t} = \int \text{cosec}^2 t\, dt$

$$= -\cot t + c$$

$$= -\cot(e^x \,.\, x) + c$$

Q. **(xii)** $\int \left(27e^{9x} + e^{12x}\right)^{1/3}$

Sol. $\int e^{3x} \left(27 + e^{3x}\right)^{1/3} dx$

Let $e^{3x} + 27 = t$

$$\Rightarrow \frac{1}{3}\int t^{1/3}\, dt$$

$$= \frac{t^{4/3}}{4} + c$$

$$= \frac{\left(e^{3x} + 27\right)^{4/3}}{4} + c$$

Q. **(xiii)** $\int \tan(10x)\tan(7x)(\tan 3x)\,dx$

Sol. $\int(\tan 10x - \tan 7x - \tan 3x)\,dx$

($\because \tan(x+y)\tan x \tan y = \tan(x+y) - \tan x - \tan y$)

$$= \frac{\ln|\sec 10x|}{10} - \frac{\ln|\sec 7x|}{7} - \frac{\ln|\sec 3x|}{3} + c$$

General Substitution

1. $\sqrt{a^2 - x^2}$ → $x = a\sin\theta$
2. $\sqrt{a^2 + x^2}$ → $x = a\tan\theta$
3. $\sqrt{x^2 - a^2}$ → $x = a\sec\theta$
4. $\sqrt{\frac{a^2 - x^2}{a^2 + x^2}}$ → $x^2 = a^2\cos 2\theta$

Using the above substitution greatly simplifies the integration as we will see in later problems.

Standard integration

13. (a) $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln\left(x + \sqrt{x^2 + a^2}\right)$

(b) $\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln\left(x + \sqrt{x^2 - a^2}\right)$

Proof: $\int \frac{dx}{\sqrt{x^2 + a^2}}$

Let $x = a\tan\theta$

$\Rightarrow dx = a\sec^2\theta\, d\theta$

$$\Rightarrow \quad \int \frac{a\sec^2\theta\, d\theta}{\sqrt{a^2\left(1+\tan^2\theta\right)}}$$

$$= \int \sec\theta\, d\theta$$

$$= \ln(\sec\theta + \tan\theta) + c$$

$$= \ln\left(\frac{\sqrt{x^2+a^2}}{a} + \frac{x}{a}\right) + c$$

$$= \ln\left(x + \sqrt{x^2+a^2}\right) + c_1 \text{ Hence Proved}$$

Similarly, for $\int \frac{dx}{\sqrt{x^2-a^2}}$, put $x = a\sec\theta$ and solve.

Ex. $\int \frac{\sin 2x}{\sqrt{9-\sin^4 x}} dx$

Sol. Put $\sin^2 x = t \Rightarrow \sin 2x\, dx = dt$

Integral becomes $\int \frac{dt}{\sqrt{9-t^2}} = \sin^{-1}\left(\frac{t}{3}\right) + c = \sin^{-1}\left(\frac{\sin^2 x}{3}\right) + c$

Ex. $\int \frac{e^x dx}{\sqrt{e^{2x}-1}}$

Sol. Put $e^x = t \Rightarrow e^x\, dx = dt$

Integral becomes $\int \frac{dt}{\sqrt{t^2-1}} = \ln\left(t + \sqrt{t^2-1}\right) + c = \ln\left(e^x + \sqrt{e^{2x}-1}\right) + c$

Ex. $\int \frac{e^x}{4+e^{2x}} dx$

Sol. Put $e^x = t$ to get $\int \frac{dt}{4+t^2} = \frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) + c = \frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + c$

Q. **Solve the following :**

(i) $\int \sqrt{\frac{x}{a-x}} dx$

Sol. $x = a\sin^2\theta$

$\Rightarrow \quad dx = a\sin 2\theta\, d\theta$

$\Rightarrow \quad \int \sqrt{\frac{a\sin^2\theta}{a\cos^2\theta}} a\sin 2\theta\, d\theta$

$$= \int a\frac{\sin\theta}{\cos\theta} \times 2\sin\theta\cos\theta\, d\theta$$

$$= a\int(1-\cos 2\theta)\, d\theta$$

$$= a\left(\theta - \frac{\sin 2\theta}{2}\right) + c$$

$$= a\left(\sin^{-1}\sqrt{\frac{x}{a}} - \frac{\sqrt{x(a-x)}}{a}\right) + c$$

Note :

In case of $\int\sqrt{\frac{x-\alpha}{\beta-x}}dx$,

put $x = \alpha\cos^2\theta + \beta\sin^2\theta$. we get,

$$\Rightarrow \int\sqrt{\frac{(\beta-\alpha)\sin^2\theta}{(\beta-\alpha)\cos^2\theta}}2(\beta-\alpha)\sin\theta\cos\theta\, d\theta$$

$$\Rightarrow \int(\beta-\alpha)2\sin^2\theta\, d\theta$$

Now we can solve it.

Q. **(ii)** $\int\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$

Sol. Let $x = \cos 2\theta$

$$\Rightarrow \int\cos\left(2\cot^{-1}(\tan\theta)\right)(-2\sin 2\theta)\, d\theta$$

$$\Rightarrow \int\cos\left(\pi - 2\tan^{-1}\tan\theta\right)(-2\sin 2\theta)\, d\theta$$

$$\Rightarrow \int 2\cos 2\theta\sin 2\theta d\theta$$

$$\Rightarrow \int\sin 4\theta\, d\theta$$

$$\Rightarrow -\frac{\cos 4\theta}{4} + c$$

$$= \frac{1-2x^2}{4} + c$$

$$= -\frac{x^2}{2} + c_1$$

Q. (iii) $\int \frac{\sqrt{\left(9-x^2\right)^3}}{x^6}\,dx$

Sol. $x = 3\sin\theta$

$$\Rightarrow \int \frac{27\cos^3\theta}{3^6\left(\sin\theta\right)^6} 3\cos\theta\, d\theta$$

$$= \int \frac{\left(\cot\theta\right)^4 \operatorname{cosec}^2\theta}{9} d\theta$$

$$= \int \frac{-t^4}{9} dt \qquad (\because \text{ put } \cot\theta = t)$$

$$= \frac{-\left(\cot\theta\right)^5}{45} + c$$

$$= -\frac{\left(\sqrt{9-x^2}\right)^5}{45x^5} + c$$

Q. (iv) $\int \frac{\sqrt{x}dx}{\sqrt{a^3-x^3}}$

Sol. Let $x^{3/2} = t$

$$\Rightarrow \frac{3}{2}\sqrt{x}\, dx = dt$$

$$\Rightarrow \frac{2}{3}\int \frac{dt}{\sqrt{a^3-t^2}}$$

$$= \frac{2}{3}\sin^{-1}\left(\frac{t}{a^{3/2}}\right) + c$$

$$= \frac{2}{3}\sin^{-1}\left(\frac{x}{a}\right)^{3/2} + c$$

Q. (v) $\int \frac{xdx}{x^4+x^2+1}$

Sol. Let $x^2 = t$

$$\Rightarrow \frac{1}{2}\int \frac{dt}{t^2+t+1}$$

$$= \frac{1}{2}\int \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{1}{2} \times \frac{2}{\sqrt{3}} \times \tan^{-1}\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + c$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2t+1}{\sqrt{3}}\right) + c$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + c$$

"FUNDA KYA HAI?"

→ For integration of type :

$$\int \frac{dx}{ax^2+bx+c} \text{ or } \int \frac{dx}{\sqrt{ax^2+bx+c}}$$

→ Make $ax^2 + bx + c$ a perfect square.

→ For integration of type :

$$\int \frac{(px+q)}{ax^2+bx+c} dx \text{ and } \int \frac{(px+q)}{\sqrt{ax^2+bx+c}} dx$$

→ Write $px + q = A\frac{d}{dx}\left(ax^2+bx+c\right) + B$

→ Now Find A and B.

→ Integration breaks into two simple parts

$\int \frac{Af'(x)}{f(x)} dx + \int \frac{B}{f(x)} dx$. Now solve it.

Ex. $\int \frac{e^x \, dx}{\sqrt{5 - 4e^x + e^{2x}}}$

Sol. Put $e^x = t$ to get $\int \frac{dt}{\sqrt{5-4t+t^2}} = \int \frac{dt}{\sqrt{(t-2)^2+1}} = \ln\left(t - 2 + \sqrt{t^2 - 4t + 5}\right) + c$,

where $t = e^x$.

Ex. $\int \frac{4x+3}{3x^2+3x+1} dx$

Sol. $4x + 3 = A(6x + 3) + B$ by equating coefficients

$A = \frac{2}{3}$ and $B = 1$

$$\Rightarrow \quad \int \frac{4x+3}{3x^2+3x+1}dx = \frac{2}{3}\int \frac{(6x+3)dx}{3x^2+3x+1} + \int \frac{dx}{3x^2+3x+1}$$

$$= \frac{2}{3}\ln\left(3x^2+3x+1\right) + \frac{1}{3}\int \frac{dx}{\left(x+\frac{1}{2}\right)^2 + \frac{1}{12}}$$

$$= \frac{2}{3}\ln\left(3x^2+3x+1\right) + \frac{1}{3}\frac{1}{\sqrt{\frac{1}{12}}}\tan^{-1}\left(\frac{x+\frac{1}{2}}{\frac{1}{\sqrt{12}}}\right) + C$$

Ex. $\int \frac{5x+4}{\sqrt{x^2+2x+5}}dx$

Sol. Let $I = \int \frac{5x+4}{\sqrt{x^2+2x+5}}dx$

let $5x + 4 = \lambda (2x + 2) + \mu$

Comparing the coefficient's, we have

$2\lambda = 5$ and $2\lambda + \mu = 4$ gives $\lambda = \frac{5}{2}$ and $\mu = -1$

Hence, we have

$$I = \frac{5}{2}\int \frac{2x+2}{\sqrt{x^2+2x+5}}dx - \int \frac{dx}{\sqrt{x^2+2x+5}} = 5\sqrt{x^2+2x+5} - \int \frac{dx}{\sqrt{(x+1)^2+2^2}}$$

$$= 5\sqrt{x^2+2x+5} - \ln\left|x+1+\sqrt{x^2+2x+5}\right| + c$$

Q. **Solve the following :**

(i) $\int \frac{(2x+3)}{x^2+2x+2}dx$

Sol. **Step-I:** $2x + 3 = A(2x + 2) + B$

$\Rightarrow A = 1, B = 1$

Step-II: $\int \frac{1(2x+2)}{x^2+2x+2}dx + \int \frac{1}{x^2+2x+2}dx$

$= \ln (x^2 + 2x + 2) + \int \frac{dx}{(x+1)^2+1}$

$= \ln(x^2 + 2x + 2) + \tan^{-1}(x + 1) + c$

Q. (ii) $\int \frac{(8x+5)}{\sqrt{4x^2+4x+2}}dx$

Sol. $8x + 5 = A(8x + 4) + B$

$\Rightarrow A = 1, B = 1$

$$\Rightarrow \int \frac{8x+4}{\sqrt{4x^2+4x+2}}dx + \int \frac{dx}{\sqrt{(2x+1)^2+1}}$$

$$\Rightarrow 2\sqrt{4x^2+4x+2} + \frac{1}{2}\ln\left((2x+1)+\sqrt{(2x+1)^2+1}\right)+c$$

Q. (iii) $\int\left(\frac{2\sin 2\theta - \cos\theta}{6-\cos^2\theta-4\sin\theta}\right)d\theta$

Sol. $$\int \frac{2(2\sin\theta\cos\theta)-\cos\theta}{6-(1-\sin^2\theta)-4\sin\theta}d\theta$$

$$= \int \frac{(4\sin\theta-1)\cos\theta\, d\theta}{\sin^2\theta-4\sin\theta+5}$$

Let $\sin\theta = y$

$$\Rightarrow \int \frac{4y-1}{y^2-4y+5}dy$$

$$= \int \frac{2(2y-4)}{y^2-4y+5}dy + \int \frac{7dy}{(y-2)^2+1}$$

$= 2\ln(y^2 - 4y + 5) + 7\tan^{-1}(y-2) + c$

$= 2\ln(\sin^2\theta - 4\sin\theta+5)+7\tan^{-1}(\sin\theta-2) + C$

Q. (iv) $\int \frac{4y-1}{y^2-4y-5}dy$

Sol. $$\int \frac{2(2y-4)+7}{y^2-4y-5}dy$$

$$= 2\ln(y^2 - 4y - 5) + 7\int \frac{dy}{(y-2)^2-9}$$

$$= 2\ln(y^2 - 4y - 5) + \frac{7}{2\times 3}\ln\left|\frac{y-2-3}{y-2+3}\right| + c$$

$$= 2\ln\left(y^2 - 4y - 5\right) + \frac{7}{6}\ln\left|\frac{y-5}{y+1}\right| + c$$

Trigonometric Function

$\int \sin^m x \cos^n x \, dx \quad ; \quad m, n \in N$

1. If one of m, n is odd and other even, then substitute term of even power.
2. If both are odd, substitute higher power Trigonometric function.
3. If both are even, use trigonometric identities to manipulate.
4. If m & n are rational numbers and 'm + n' is a 'negative even' like {-2, -4, -6,...}, then substitute tan x = t or cot x = t.

Ex. $\int \sin^5 x \cos^6 x \, dx$

Sol. $6 > 5 \Rightarrow \cos x = t$

$$\Rightarrow \int (\sin x)^4 (\cos x)^6 \sin x dx$$

$$= -\int (1-t^2)^2 t^6 dt$$

$$= \int (-t^{10} + 2t^8 - t^6) dt$$

$$= \frac{-t^{11}}{11} + \frac{2t^9}{9} - \frac{t^7}{7} + C = \frac{-(\cos x)^{11}}{11} + \frac{2\cos^9 x}{9} - \frac{\cos^7 x}{7} + c$$

Ex. $\int \sin^3 x \cos^5 x \, dx$

Sol. Both power are odd and 5 > 3. So, cos x = t

$$\int \sin^2 x \cos^5 x (\sin x dx)$$

$$= -\int (1-t^2) t^5 dt$$

$$= \frac{-t^6}{6} + \frac{t^8}{8} + C$$

$$= -\frac{\cos^6 x}{6} + \frac{\cos^8 x}{8} + C$$

Ex. $\int \frac{dx}{\sin x \cos^3 x}$

Sol. $m = -1, n = -3, m + n = -4$

$\therefore$ let $\tan x = t$

$$\int \frac{\sec^4 x}{\tan x} dx$$

$$= \int \frac{\sec^2 x}{\tan x} \left(\sec^2 x \, dx\right)$$

$$= \int \left(\frac{1+t^2}{t}\right) dt$$

$$= \ln|t| + \frac{t^2}{2} + C$$

$$= \ell n|\tan x| + \frac{\tan^2 x}{2} + C$$

Ex. $\int \sin^2 x \cos^4 x \, dx$

Sol. $\int (\sin x \cos x)^2 \cos^2 x \, dx$

$$= \int \left(\frac{\sin 2x}{2}\right)^2 \left(\frac{1+\cos 2x}{2}\right) dx$$

$$= \frac{1}{8} \int \left(\frac{1-\cos 4x}{2}\right)(1+\cos 2x) dx$$

$$= \frac{1}{16} \int (1 + \cos 2x - \cos 4x - \cos 2x \cos 4x) \, dx$$

$$= \frac{1}{16} \left(x + \frac{\sin 2x}{2} - \frac{\sin 4x}{4} - \frac{\sin 6x}{12} - \frac{\sin 2x}{4} + C\right)$$

Ex. **Solve the following :** $\int \frac{dx}{\sin^{\frac{7}{2}} x \cos^{\frac{13}{2}} x}$

Sol. $m = -\frac{7}{2}, n = -\frac{13}{2}, m + n = -10$

$$\int \frac{1}{\sin x^{\frac{7}{2}} \cos^{\frac{13}{2}} x} \times \frac{(\cos x)^{\frac{7}{2}}}{(\cos x)^{\frac{7}{2}}} dx$$

$$\int (\cot x)^{\frac{7}{2}} (\sec x)^{10} dx$$

Put $\tan x = t$

$$\Rightarrow \quad \int \frac{(1+t^2)^4}{t^{\frac{7}{2}}} dt = \int \left(t^{-\frac{7}{2}} + 4t^{-\frac{3}{2}} + 6\sqrt{t} + 4t^{\frac{5}{2}} + t^{\frac{9}{2}} \right) dt$$

$$= \frac{-2}{5}(\tan x)^{-\frac{5}{2}} - \frac{8}{\sqrt{\tan x}} + 4(\tan x)^{\frac{3}{2}} + \frac{8}{7}(\tan x)^{\frac{7}{2}} + \frac{2(\tan x)^{\frac{11}{2}}}{11} + C$$

INTEGRATION BY PARTS:

Theory: If f(x) and g(x) are derivable functions then

$$\frac{d}{dx}\left[f(x).g(x)\right] = f(x).g'(x) + g(x).f'(x)$$

$$\therefore \int \underset{I}{f(x)}.\underset{II}{g'(x)}dx = f(x).g(x) - \int g(x).f'(x)dx$$

$$I = \int \underbrace{f(x)}_{I}.\underbrace{g(x)}_{II} dx$$

$$= 1^{st} \text{ function} \times \text{integeral } 2^{nd} - \int (\text{diff.coeff. of } 1^{st}) \times (\text{integral of } 2^{nd}) dx$$

Remember ILATE for deciding the choice of the first and second function which is arbitrary.

Here I for inverse trigonometric function

L for Logarithmic function

A for Algebraic function

T for Trigonometric function

E for Exponential function

Selection of Ist and IInd Function:

(i) Choose 2nd function which is easily integrable (Use ILATE)

(ii) Choose 1st and 2nd function such that after by-parts, complexity of 2nd term reduces as compared to original integration.

(iii) Note that sometimes, '1' is also taken as function.

Ex. $\int \underbrace{(\ell nx)}_{\text{log}} \underbrace{x}_{\text{Algebra}} dx$

Sol. According to ILATE 'ln x' will be 1st function and 'x' will be 2nd function

$$\Rightarrow \int x\ell n(x)dx = \frac{x^2}{2}\ell n(x) - \int\left(\frac{x^2}{2}\right)\times\left(\frac{1}{x}\right)dx$$

$$= \frac{x^2}{2}\ell n(x) - \frac{x^2}{4} + C$$

Ex. $\int x\cos x\,dx$

Sol. Take x as first and cos x as 2nd function.

$$\int x\cos x\,dx = x\sin x - \int 1\times\sin x\,dx = x\sin x + \cos x + c$$

Ex. $\int x\tan^{-1}x dx$

Sol. $\int \underset{\text{II}}{\underset{\sim}{x}}\underbrace{\tan^{-1}x\,dx}_{\text{I}} = \left(\tan^{-1}x\right)\frac{x^2}{2} - \int\frac{x^2}{2\left(1+x^2\right)}dx$

$$= \frac{1}{2}x^2\left(\tan^{-1}x\right) - \frac{1}{2}\int\frac{1+x^2-1}{1+x^2}dx$$

$$= \frac{1}{2}x^2\tan^{-1}x - \frac{x}{2} + \frac{\tan^{-1}x}{2} + c$$

Ex. $\int\frac{\sin^{-1}x}{\left(1-x^2\right)^{\frac{3}{2}}}dx$

Sol. Put $\sin^{-1}x = \theta$ or $x = \sin\theta \Rightarrow dx = \cos\theta d\theta$

Integral becomes $\int\frac{\theta.\cos\theta\,d\theta}{\cos^3\theta} = \int\underset{\text{I}}{\underset{\sim}{\theta}}.\underbrace{\sec^2\theta\,d\theta}_{\text{II}}$

$$= \theta.\tan\theta - \int 1.\tan\theta d\theta = \theta\tan\theta - \ln\left(\sec\theta\right) + c = \left(\sin^{-1}x\right).\frac{x}{\sqrt{1-x^2}} + \ln\left(\sqrt{1-x^2}\right) + c$$

Ex. $\int \sin(\ln x)\,dx$

Sol. Put $\ln x = t$ to get $I = \int \underset{I}{e^t}\ \underset{II}{\sin t}\,dt = e^t(-\cos t) - \int e^t(-\cos t)\,dt$

$= -\cos t.e^t + \int e^t \cos t\,dt = -\cos t.e^t + \left[e^t \sin t - \int e^t \sin t\,dt\right]$

$I = -\cos t.e^t + e^t \sin t - I \quad \text{or} \quad I = \frac{1}{2}e^t(\sin t - \cos t) + c$

$= \frac{1}{2}x(\sin(\ln x) - \cos(\ln x)) + c$

Ex. $\int x^2 e^{3x}dx$

Sol. $I = \int \underset{I}{x^2}\ \underset{II}{e^{3x}}\,dx = \frac{x^2e^{3x}}{3} - \int 2x.\frac{e^{3x}}{3}dx$

$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\int \underset{I}{x}\ \underset{II}{e^{3x}}\,dx = \frac{1}{3}x^2e^{3x} - \frac{2}{3}\left[x.\frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}dx\right]$

$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{27}e^{3x} + c$

Q. **Solve the function :**

(i) $\int xe^x dx$

Sol. $\int \underset{A}{x}\ \underset{E}{e^x}\,dx$

$\therefore$ $x \to 1^{st}$ function

$e^x \to 2^{nd}$ function

$\Rightarrow$ $xe^x - \int 1.e^x dx$

$= xe^x - e^x + c$

Q. **(ii)** $\int \ell n(x)dx$

Sol. $\int \underbrace{(1)}_{A}.\underbrace{(\ell n(x))}_{L}dx$

$\Rightarrow$ $\ln(x) \to 1^{st}$ function; $1 \to 2^{nd}$ function

$\Rightarrow$ $x\ell n(x) - \int \frac{1}{x}.x\,dx$

$= x\ell n(x) - x + c$

Q. **(iii)** $\int (\ell nx)^2 \, dx$

Sol. $(\ell nx)^2 \rightarrow 1^{st}$ function; $1 \rightarrow 2^{nd}$ fnction

$$\Rightarrow \quad (\ell n(x))^2 .x - \int (2\ell nx).\left(\frac{1}{x}\right).x \, dx$$

$$= x(\ell n(x))^2 - 2\int \ell n(x) dx$$

$$= x(\ell n(x))^2 - 2x\,\ell n(x) + 2x + c$$

Q. **(iv)** $\int \sin^{-1} x \, dx$

Sol. $\sin^{-1} x \rightarrow 1^{st}$ function; $1 \rightarrow 2^{nd}$ function

$$\Rightarrow \quad x \sin^{-1} x - \int \left(\frac{1}{\sqrt{1-x^2}}\right) x \, dx$$

Let $1 - x^2 = t$

$\Rightarrow -2x \, dx = dt$

$$\Rightarrow \quad \int \frac{-x dx}{\sqrt{1-x^2}} = \int \frac{dt}{2\sqrt{t}} = \sqrt{t} + c = \sqrt{1-x^2} + c$$

$$\Rightarrow \quad \int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1-x^2} + c$$

Q. **(v)** $\int x^2 \ell n(x) dx$

Sol. $\ell n(x) \rightarrow 1^{st}$ function; $x^2 \rightarrow 2^{nd}$ function

$$\Rightarrow \quad \frac{x^3}{3} \ell n(x) - \int \left(\frac{1}{x}\right)\left(\frac{x^3}{3}\right) dx$$

$$= \frac{x^3}{3} \ell n(x) - \frac{x^3}{9} + c$$

Q. **(vi)** $\int x \cos x \sin^2 x \, dx$

Sol. We can observe that $\cos x \sin^2 x$ can be easily integrated

So, $\cos x \sin^2 x \rightarrow 2^{nd}$ function; $x \rightarrow 1^{st}$ function.

$$\Rightarrow x \int \sin^2 x \cos x dx - \int (1)\left(\int \sin^2 x \cos x \, dx\right) dx$$

If $\sin x = t$

$$\int \sin^2 x \cos x \, dx = \int t^2 dt = \frac{t^3}{3} = \frac{\sin^3 x}{3}$$

$$\therefore \int x \cos x \sin^2 x dx = x\left(\frac{\sin^3 x}{3}\right) - \int \frac{\sin^3 x}{3} dx$$

$$= \frac{x \sin^3 x}{3} - \int\left(\frac{1-\cos^2 x}{3}\right)\sin x \, dx$$

$$= \frac{x \sin^3 x}{3} + \frac{\cos x}{3} - \frac{\cos^3 x}{9} + c$$

Q. **(vii)** $\int \theta \tan\theta \sec^2\theta \, d\theta$

Sol. $\int \sec^2\theta \tan\theta \, d\theta = \frac{\tan^2\theta}{2}$

$$\therefore \int \theta\left(\tan\theta \sec^2\theta\right) d\theta = \frac{\theta \tan^2\theta}{2} - \int \frac{\tan^2\theta}{2} d\theta$$

$$= \frac{\theta \tan^2\theta}{2} - \frac{1}{2}\int\left(\sec^2\theta - 1\right) d\theta$$

$$= \frac{\theta \tan^2\theta}{2} - \frac{1}{2}\left(\tan\theta - \theta\right) + c$$

$$= \frac{\theta \tan^2\theta - \tan\theta + \theta}{2} + c$$

Standard Integration

1. $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2}\left(a \sin bx - b \cos bx\right) + c$

Proof :

Let $I = \int e^{ax} \sin bx \, dx$

$$\Rightarrow I = \sin bx\left(\frac{e^{ax}}{a}\right) - \int b \cos bx\left(\frac{e^{ax}}{a}\right) dx$$

$$\Rightarrow I = \frac{\sin bx \, e^{ax}}{a} - \frac{b}{a}\left\{\cos bx \frac{e^{ax}}{a} - \int\left(-b \sin bx\right)\frac{e^{ax}}{a} dx\right\}$$

$$\Rightarrow I = \left(\frac{a \sin bx - b \cos bx}{a^2}\right)e^{ax} - \frac{b^2}{a^2} I + c$$

$$\Rightarrow \frac{I\left(a^2+b^2\right)}{a^2}=\left(\frac{a\sin bx-b\cos bx}{a^2}\right)e^{ax}+c$$

$$\Rightarrow I=\left(\frac{a\sin bx-b\cos bx}{a^2+b^2}\right)e^{ax}+c \text{ Hence proved}$$

2. $\int e^{ax}\cos bx\,dx = e^{ax}\dfrac{(a\cos bx+b\sin bx)}{a^2+b^2}+c$

Proof :

Let $\int e^{ax}\cos bx\,dx = e^{ax}\{A\cos bx+B\sin bx\}+c$

Differentiating both side, we get :

$e^{ax}\cos bx = e^{ax}\{a(A\cos bx+B\sin bx)+(-Ab\sin bx+Bb\cos bx)\}$

Equating the coefficients,

$1 = aA + Bb$

$0 = aB - bA$

$\Rightarrow B=\dfrac{b}{a^2+b^2}; A=\dfrac{a}{a^2+b^2}$ **Hence Proved.**

Ex. $\int e^x \sin x dx$

Sol. $a = 1, b = 1.$

$$\therefore \int e^x \sin x\,dx = e^x\left(\frac{\sin x-\cos x}{1^2+1^2}\right)+c$$

$$= e^x\left(\frac{\sin x-\cos x}{2}\right)+c$$

Ex. $\int e^x \cos x\,dx$

Sol. $a = 1,\ b = 1$

$$\Rightarrow \int e^x \cos x\,dx = e^x\left(\frac{\cos x+\sin x}{2}\right)$$

Q. **Solve the following :**

(i) $\int \operatorname{cosec}^2 x \ln(\sec x)\,dx$

Sol. $\int \operatorname{cosec}^2 x \ln(\sec x)\,dx$

$= (-\cot x)\,\ell n(\sec x) + \int \cot x \tan x\,dx$

$= -\cot x\,\ell n(\sec x) + x + c$

Q. **(ii)** $\int \cos^{-1}\left(\frac{1}{x}\right)dx$

Sol. $\int \cos^{-1}\left(\frac{1}{x}\right)dx$

$= \left(\sec^{-1} x\right)(x) - \int\left(\frac{1}{x\sqrt{x^2-1}}\right)x\,dx$

$= \sec^{-1} x - \ell n\left(x + \sqrt{x^2-1}\right) + c$

Q. **(iii)** $\int \sin x\,\ell n(\sec x + \tan x)\,dx$

Sol. $\sin x \Rightarrow 2^{nd}$ function $\quad \ell n(\sec x + \tan x) \rightarrow 1^{st}$ function

$\Rightarrow \left(\int \sin x\,dx\right)\ell n(\sec x + \tan x) - \int \sec x\left(\int \sin x\,dx\right)dx$

$= -\cos x\,\ell n(\sec x + \tan x) + \int \sec x \cos x\,dx$

$= -\cos x\,\ell n(\sec x + \tan x) + x + c$

Q. **(iv)** $\int \ell n\left(x + \sqrt{x^2+a^2}\right)dx$

Sol. Applying by parts, we get

$x\ell n\left(x + \sqrt{x^2+a^2}\right) - \int x \cdot \frac{1}{\sqrt{x^2+a^2}}\,dx$

$= x\ell n\left(x + \sqrt{x^2+a^2}\right) - \int \frac{2x}{2\sqrt{x^2+a^2}}\,dx$

$= x\ell n\left(x + \sqrt{x^2+a^2}\right) - \sqrt{x^2+a^2} + c$

Q. **(v)** $\int x\cos x\cos 2x dx$

Sol. $\int \frac{x}{2}(\cos 3x + \cos x)dx$

$$= \frac{x}{2}\left(\frac{\sin 3x}{3} + \sin x\right) - \frac{1}{2}\int\left(\frac{\sin 3x}{3} + \sin x\right)dx$$

$$= \frac{x\sin 3x}{6} + \frac{x\sin x}{2} + \frac{1}{2}\left(\frac{\cos 3x}{9} + \cos x\right) + c$$

$$= \frac{x\sin 3x}{6} + \frac{x\sin x}{2} + \frac{\cos 3x}{18} + \frac{\cos x}{2} + c$$

Q. **(vi)** $\int e^x(1+x)\ell n\left(xe^x\right)dx$

Sol. Let $xe^x = y$

$\Rightarrow e^x(1+x)dx = dy.$

$\Rightarrow \int \ell n y dy$

$= y\ell ny - y + c$

$= xe^x\ell n\left(xe^x\right) - xe^x + c$

Q. **(vii)** $\int \sin^{-1}\sqrt{\frac{x}{a+x}}dx$

Sol. Let $\sin^{-1}\sqrt{\frac{x}{a+x}} = \theta$ (Determining substitution)

$\Rightarrow \sin^2\theta = \frac{x}{a+x}$

$\Rightarrow a\sin^2\theta = x\cos^2\theta$

$\Rightarrow x = a\tan^2\theta$

$\Rightarrow dx = 2a\tan\theta\sec^2\theta d\theta$

Now using it, we get

$\int \theta\left(2a\tan\theta\sec^2\theta\right)d\theta$

$= \theta\left(a\tan^2\theta\right) - \int 1.\left(a\tan^2\theta\right)d\theta$

$= a\theta\tan^2\theta - a\int\left(\sec^2\theta - 1\right)d\theta$

$= a\theta\tan^2\theta - a\tan\theta + a\theta + c$

Q. **(viii)** $\int \frac{\cos^{-1} x}{x^3} dx$

Sol. $\left(\cos^{-1} x\right).\left(\frac{-1}{2x^2}\right) + \int\left(\frac{-1}{\sqrt{1-x^2}}\right)\left(\frac{1}{2x^2}\right) dx$

$$= \frac{-\cos^{-1} x}{2x^2} - \int \frac{1}{2x^3\sqrt{\frac{1}{x^2}-1}} dx$$

Let $\frac{1}{x^2} - 1 = y^2$

$\Rightarrow \frac{-2}{x^3} dx = 2y\, dy$

$\Rightarrow \frac{dx}{x^3} = -y dy$

$$= \frac{-1}{2}\sqrt{\frac{1}{x^2}-1} + c$$

$$= \frac{-\cos^{-1} x}{2x^2} + \frac{1}{2}\sqrt{\frac{1}{x^2}-1} + c$$

Q. **(ix)** $\int \ell n\left(1+x\right)^{1+x} dx$

Sol. $\int (1+x)\,\ell n(1+x)\, dx$

$$= \left(x + \frac{x^2}{2}\right)\ell n(1+x) - \int\left(x + \frac{x^2}{2}\right)\frac{1}{(1+x)} dx$$

$$= \frac{\left(x^2+2x\right)\ell n(1+x)}{2} - \frac{1}{2}\int\left(\frac{x^2+2x+1-1}{1+x}\right) dx$$

$$= \frac{\left(x^2+2x\right)\ell n(1+x)}{2} - \frac{1}{2}\left(\frac{x^2}{2} + x - \ell n(1+x)\right) + c$$

$$= \frac{(x+1)^2\,\ell n(1+x)}{2} - \frac{x^2}{4} - \frac{x}{2} + c$$

Q. **(x)** $\int \frac{x\,dx}{1+\sin x}dx$

Sol. $\int \frac{x(1-\sin x)}{\cos^2 x}dx$

$= \int x\sec^2 x\,dx - \int x\sec x\tan x\,dx$

$= \left(x\tan x - \int \tan x\,dx\right) - \left(x\sec x - \int \sec x\,dx\right)$

$= x\tan x - \ell n(\sec x) - x\sec x + \ell n(\sec x + \tan x) + c$

Q. **(xi)** $\int \ell nx.\frac{1}{(x+1)^2}dx$

Sol. Applying by parts,

$-\frac{\ell nx}{x+1} + \int \frac{1}{x(x+1)}dx$

$= \frac{-\ell nx}{x+1} + \int\left(\frac{1}{x} - \frac{1}{x+1}\right)dx$

$= -\frac{\ell nx}{x+1} + \ell nx - \ell n(x+1) + c$

$= -\frac{\ell nx}{x+1} + \ell n\left(\frac{x}{x+1}\right) + c$

Q. **(xii)** $\int x\sin^{-1}\left(\frac{1}{2}\sqrt{\frac{2a-x}{a}}\right)dx$

Sol. Let $x = 2a\cos 2t$

$dx = -4a\sin 2t dt$

$\Rightarrow \int 2a\cos 2t\sin^{-1}\left(\frac{1}{2}\sqrt{\frac{2a(1-\cos 2t)}{a}}\right)(-4a\sin 2t)\,dt$

$= \int -8a^2\cos 2t\sin 2t\sin^{-1}(\sin t)\,dt$

$= \int -4a^2\sin(4t)\,tdt$

$= -4a^2\left(\frac{-\cos 4t}{4}\right)t + 4a^2\int 1.\left(\frac{-\cos 4t}{4}\right)at$

$= a^2 t\cos 4t - a^2\frac{\sin 4t}{4} + c$

Standard Integration :

1. $\int \sqrt{a^2 + x^2}\,dx = \frac{x}{2}\sqrt{a^2 + x^2} + \frac{a^2}{2}\ell n\left[x + \sqrt{x^2 + a^2}\right] + c$

Proof :

$x = a\tan t$

$dx = a\sec^2 t\,dt$

$\Rightarrow \quad I = \int \sqrt{a^2 + x^2}\,dx = \int a^2 \sec^3 t\,dt$

$\Rightarrow \quad I = a^2 \int (\sec^2 t)(\sec t)\,dt$

$= a^2 \tan t \sec t - a^2 \int \sec t \tan^2 t\,dt$

$= a^2 \tan t \sec t - a^2 \int (\sec^3 t - \sec t)\,dt$

$\Rightarrow \quad I = a^2 \tan t \sec t - I + a^2 \int \sec t\,dt$

$\Rightarrow \quad I = \frac{a^2}{2}\left(\frac{\sqrt{x^2 + a^2}}{a}\right).\left(\frac{x}{a}\right) + \frac{a^2}{2}\ell n\left(\frac{\sqrt{x^2 + a^2}}{a} + \frac{x}{a}\right) + c$

$\Rightarrow \quad I = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ell n\left(\sqrt{x^2 + a^2} + x\right) + c$ **Hence Proved.**

2. $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$

Proof:

$I = \int \sqrt{a^2 - x^2}\,dx = x\sqrt{a^2 - x^2} - \int x\left(\frac{1}{2\sqrt{a^2 - x^2}}\right)(-2x)\,dx$

$\Rightarrow \quad I = x\sqrt{a^2 - x^2} - \int \frac{a^2 - x^2 - a^2}{\sqrt{a^2 - x^2}}\,dx$

$\Rightarrow \quad I = x\sqrt{a^2 - x^2} - I + a^2 \int \frac{dx}{\sqrt{a^2 - x^2}}$

$\Rightarrow \quad 2I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) + c$

$\Rightarrow \quad I = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$ **Hence Proved.**

3. $\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ell n\left(\sqrt{x^2 - a^2} + x\right) + c$

Proof :

Let $x = a\sec\theta$

Now, similar to above identities, apply by parts and prove it.

Two Classic Integrands :

(a) $\int e^x\left(f(x) + f'(x)\right)dx = e^x f(x) + C$

Proof :

$$\int e^x\left(f(x) + f'(x)\right)dx = \int e^x f(x)\,dx + \int \underset{I}{e^x}\ \underset{II}{f'(x)}\,dx$$

$$= \int e^x f(x)\,dx + e^x f(x) - \int e^x f(x)\,dx + c = e^x f(x) + c$$

(b) $\int\left(f(x) + xf'(x)\right)dx = x\,f(x) + C$

Proof :

$$\int\left(f(x) + xf'(x)\right)dx$$

$$= \int f(x)\,dx + \int \underset{I}{x}\,\underbrace{f'(x)}_{II}\,dx$$

$$= \int f(x)\,dx + x\,f(x) - \int f(x)\,dx$$

$$= x\,f(x) + c$$

Ex. $\int e^x(x+1)\,dx$

Sol. $\int e^x(x+1)\,dx$

$\downarrow\ \downarrow$

$f(x)\ \ f'(x)$

$\Rightarrow\ e^x f(x) + c$

$\Rightarrow\ e^x x + c$

Ex. $\int \frac{xe^x}{(1+x)^2}dx$

Sol. $\int \frac{xe^x}{(1+x)^2}dx = \int e^x \frac{1+x-1}{(1+x)^2}dx = \int e^x \left\{\frac{1}{1+x} + \frac{-1}{(1+x)^2}\right\}dx = e^x \frac{1}{1+x} + c$

Ex. $\int \frac{e^x}{x}(1+x.\ln x)dx$

Sol. $\int \frac{e^x}{x}(1+x.\ln x)dx = \int e^x \left(\frac{1}{x} + \ell nx\right) = e^x \ln x + c$

Ex. $\int \frac{e^x x^2}{(x+2)^2}dx$

Sol. $\int e^x \frac{x^2}{(x+2)^2}dx = \int e^x \frac{x^2-4+4}{(x+2)^2}dx = \int e^x \left\{\frac{x^2-4}{(x+2)^2} + \frac{4}{(x+2)^2}\right\}dx$

$\int e^x \left\{\frac{x-2}{x+2} + \frac{4}{(x+2)^2}\right\}dx = e^x \left(\frac{x-2}{x+2}\right) + c$

Ex. $\int (\sin x + x\cos x)dx$

Sol. $\int \left(\underbrace{\sin x}_{f(x)} + x\underbrace{\cos x}_{f'(x)}\right)dx = x\sin x + c$

Ex. $\int \left(2\ln x + (\ln x)^2\right)dx$

Sol. $\int \left(2\ln x + (\ln x)^2\right)dx = \int \left(x.\frac{2\ln x}{x} + (\ln x)^2\right)dx = x(\ln x)^2$

Ex. $\int \left(\ell n(\ell nx) + \frac{1}{\ell n^2 x}\right)dx$

Sol. Put $\ln x = t$ to get $\int e^t \left(\ell nt + \frac{1}{t^2}\right)dt = \int e^t \left(\ln t - \frac{1}{t} + \frac{1}{t} + \frac{1}{t^2}\right)$

$= e^t \left(\ln t - \frac{1}{t}\right) + c = e^{\ln x}\left(\ln(\ln x) - \frac{1}{\ln x}\right) + c = x\left[\ln(\ln x) - \frac{1}{\ln x}\right] + c$

Q. **Solve the following :**

(i) $\int e^x (\cos x + \sin x)\,dx$

Sol. $f(x) = \sin x, f'(x) = \cos x$

$= e^x \sin x + c$

Q. **(ii)** $\int e^x (\cos x - \sin x)\,dx$

Sol. $f(x) = \cos x \qquad f'(x) = -\sin x$

$\Rightarrow \int e^x (\cos x - \sin x)\,dx = e^x \cos x + c$

Q. **(iii)** $\int e^x (\tan x + \sec^2 x)\,dx$

Sol. $f(x) = \tan x, f'(x) = \sec^2 x$

$\Rightarrow \int e^x (\tan x + \sec^2 x)\,dx = e^x \tan x + c$

Q. **(iv)** $\int e^x \sec x(1 + \tan x)\,dx$

Sol. $I = \int e^x (\sec x + \sec x \tan x)\,dx$

$f(x) = \sec x \quad \& f'(x) = \sec x \tan x$

$\Rightarrow I = e \ \sec x + c$

Q. **(v)** $\int e^{2x} \left(\frac{\sin 4x - 2}{1 - \cos 4x}\right) dx$

Sol. Let $2x = y$

$\Rightarrow I = \frac{1}{2}\int e^y \left(\frac{\sin 2y - 2}{1 - \cos 2y}\right) dy$

$\Rightarrow = \frac{1}{2}\int e^y \left(\frac{2 \sin y \cos y - 2}{2 \sin^2 y}\right) dy$

$= \frac{1}{2}\int e^y (\cot y - \operatorname{cosec}^2 y)\,dy$

Again, it is of form $\int e^x (f(x) + f'(x))dx$

$\Rightarrow I = \frac{1}{2} e^y \cot y + c = \frac{e^{2x} \cot 2x}{2} + c$

Q. **(vi)** $\int\left(\tan x + x\sec^2 x\right)dx$

Sol. If $f(x) = \tan x \Rightarrow f'(x) = \sec^2 x$

$\therefore$ Above integral is of the form $\int\left(f(x) + xf'(x)\right)dx$

$\Rightarrow \int\left(\tan x + x\sec^2 x\right)dx = x\tan x + c$

Q. **(vii)** $\int\frac{x + \sin x}{1 + \cos x}dx$

Sol. $I = \int\left(\frac{x}{2\cos^2\frac{x}{2}} + \tan\frac{x}{2}\right)dx$

$= \int\left(x\left(\frac{\sec^2\left(\frac{x}{2}\right)}{2}\right) + \tan\frac{x}{2}\right)dx$

If $f(x) = \tan\frac{x}{2} \Rightarrow f'(x) = \frac{\sec^2\left(\frac{x}{2}\right)}{2}$

$\Rightarrow I = x\tan\left(\frac{x}{2}\right) + c$

Q. **(viii)** $\int e^x\left(\frac{x^2 + 5x + 7}{(x+3)^2}\right)dx$

Sol. $I = \int e^x\left(\frac{(x+3)^2 - (x+3) + 1}{(x+3)^2}\right)dx$

$= \int e^x dx - \int e^x\left(\frac{1}{x+3} - \frac{1}{(x+3)^2}\right)dx$

$\Rightarrow I = e^x - \frac{e^x}{x+3}dx$

Q. **(ix)** $\int e^x \left(\frac{x^2+1}{(x+1)^2} \right) dx$

Sol. $I = \int e^x \left(\frac{x^2-1}{(x+1)^2} + \frac{2}{(x+1)^2} \right) dx$

$= \int e^x \left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2} \right) dx$

If $f(x) = \frac{x-1}{x+1}, f'(x) = \frac{2}{(x+1)^2}$

$\Rightarrow \quad I = e^x \left(\frac{x-1}{x+1} \right) + c$

Q. **(x)** $\int \frac{x - \sin x}{1 - \cos x} dx$

Sol. $I = \int \left(\frac{x}{2\sin^2\left(\frac{x}{2}\right)} - \cot\frac{x}{2} \right) dx$

$= \int \left[x\left(\frac{1}{2}\operatorname{cosec}^2\frac{x}{2} \right) - \cot\frac{x}{2} \right] dx$

$\because \frac{d}{dx}\left(-\cot\frac{x}{2} \right) = \frac{1}{2}\operatorname{cosec}^2\left(\frac{x}{2}\right)$

$\Rightarrow \quad I = -x\cot\left(\frac{x}{2}\right) + c$

Q. **(xi)** $\int e^x \left(\frac{1}{(1+x^2)^{3/2}} + \frac{x}{\sqrt{1+x^2}} \right) dx$

Sol. First check the $\frac{d}{dx}\left(\frac{x}{\sqrt{1+x^2}} \right)$

$\frac{d}{dx}\left(\frac{x}{\sqrt{1+x^2}} \right) = \frac{\sqrt{1+x^2} - x\left(\frac{2x}{2\sqrt{1+x^2}} \right)}{\left(\sqrt{1+x^2} \right)^2}$

$$= \frac{1+x^2-x^2}{(1+x^2)^{\frac{3}{2}}} = \frac{1}{(1+x^2)^{\frac{3}{2}}}$$

$\Rightarrow$ Integral becomes the form of $\int e^x\left(f(x)+f'(x)\right)dx$

$\Rightarrow \quad I = \frac{xe^x}{\sqrt{1+x^2}} + c$

Q. **(xii)** $\int\left(\frac{\ell n(x)}{\left(1+\ell n(x)\right)^2}\right)dx$

Sol. $I = \int\left(\frac{1+\ell n(x)}{\left(1+\ell n(x)\right)^2} - \frac{1}{\left(1+\ell n(x)\right)^2}\right)dx$

$$= \int\left[\frac{1}{1+\ell n(x)} - x\left(\frac{1}{\left(1+\ell n(x)\right)^2}\right).\left(\frac{1}{x}\right)\right]dx$$

$\Rightarrow \quad I = \frac{x}{1+\ln(x)} + c$

Q. **(xiii)** $\int\frac{e^x(x-1)}{(x+1)^3}dx$

Sol. $\int e^x\left(\frac{x+1}{(x+1)^3} - \frac{2}{(x+1)^3}\right)dx$

$$= \int e^x\left(\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}\right)dx \quad = \frac{e^x}{(x+1)^2} + c$$

Q. **(xiv)** $\int e^x\left(x^4\right)dx$

Sol. $I = \int e^x\left[\left(x^4+4x^3\right)-\left(4x^3+12x^2\right)+\left(12x^2+24x\right)-\left(24x+24\right)+24\right]dx$

(continuous making $e^x\left(f(x)+f'(x)\right)$ from)

$\Rightarrow \quad I = e^x\left(x^4-4x^3+12x^2-24x+24\right)+c$

THE SECRET CONCEPT-MIND IT!

If f(x) is a polynomial of degree 'n' then $\int e^x f(x)dx = e^x g(x) + c$ where g(x) will also be polynomial of degree 'n'.

For eg.

$$\int e^x \left(ax^2 + bx + c\right)dx = e^x \left(px^2 + qx + n\right) + d$$

So just differentiate both side and equate coefficients to get p, q and r.

Ex. $\int e^{3x}x^2dx$

Sol. Let $\int e^{3x}x^2dx = e^{3x}\left(Ax^2 + Bx + C\right) + D$

Differentiating both side, we get

$$e^{3x}x^2 = e^{3x}\left(3Ax^2 + 3Bx + 3C + 2Ax + B\right)$$

$$\Rightarrow \quad 1 = 3A \Rightarrow A = \frac{1}{3}$$

$$\Rightarrow \quad 0 = 3B + 2A \Rightarrow B = -\frac{2}{9}$$

$$\Rightarrow \quad 0 = 3C + B \Rightarrow C = \frac{2}{27}$$

$$\therefore \quad \int e^{3x}x^2dx = e^{3x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right) + D$$

Q. **Find $\int \left(x^3 + 3x + 1\right)e^{3x}dx$**

Sol. $I = \int e^{3x}\left(x^3 + 3x + 1\right) = e^{3x}\left(Ax^3 + Bx^2 + Cx + D\right) + K$

Differentiating, we get

$$e^{3x}\left(x^3 + 3x + 1\right) = e^{3x}\left(3Ax^3 + 3Bx^2 + 3Cx + 3D + 3Ax^2 + 2Bx + C\right)$$

$$\Rightarrow \quad 1 = 3A \Rightarrow A = \frac{1}{3}$$

$$0 = 3B + 3A \Rightarrow B = -\frac{1}{3}$$

$$3 = 3C + 2B \Rightarrow C = \frac{11}{9}$$

$$1 = 3D + C \Rightarrow D = -\frac{2}{27}$$

$$\Rightarrow \quad I = e^{3x}\left(\frac{x^3}{3} - \frac{x^2}{3} + \frac{11x}{9} - \frac{2}{27}\right) + k$$

General Concept

$$\int e^{g(x)}\left(f(x)g'(x) + f'(x)\right)dx = e^{g(x)}f(x) + C$$

Ex. $\int e^{\tan x}\left(\sin x - \sec x\right)dx$

Sol. $\int e^{\tan x}\left(\sin x + (-\cos x)\sec^2 x\right)dx$

Now $\frac{d}{dx}(\tan x) = \sec^2 x$ and $\frac{d}{dx}(-\cos x) = \sin x$

$$\Rightarrow \quad I = e^{\tan x}(-\cos x) + c$$

PARTIAL FRACTION

This technique is used if a rational function is being integrated whose denominator can be factorised

If integrand is $\frac{P_1(x)}{P_2(x)}$, $P_1(x)$ & $P_2(x)$ are polynomials.

Case-I:

If degree of $P_1(x) \geq$ degree of $P_2(x)$

→ Divide and move to case II.

Case-II:

If degree of $P_1(x) < P_2(x)$

→ Now that the degree of $P_1(x)$ is less than that of $P_2(x)$, we can integrate the function after coefficient comparison. It can be understood form the following illustrations.

Ex. $\int \frac{x^2+2x+4}{(x-1)(x-2)(x+1)}dx$

Sol. $\int \frac{x^2+2x+4}{(x-1)(x-2)(x+1)} = \frac{A}{x-1}+\frac{B}{x-2}+\frac{C}{x+1}$

Now, $x^2+2x+4 = A(x-2)(x+1)+B(x-1)(x+1)+C(x-1)(x-2)$

Now using coefficient comparison of different exponents of x, we get,

$1 = A + B + C$ (coefficient of x^2)

$2 = -A - 3C$ (coefficient of x)

$4 = -2A - B + 2C$ (comparing constants)

$\Rightarrow \quad A = -\frac{7}{2};\ B = 4;\ C = \frac{1}{2}$

Alternate Method:

Put x = 1 on both side;

$\Rightarrow \quad 1^2+2\times1+4 = A(-1)(2)+B\times0+C\times0$

$\Rightarrow \quad A = -\frac{7}{2}$

Put x = 2 on both side;

$\Rightarrow \quad 2^2+2\times2+4 = A(0)+B\times3+C\times0$

$\Rightarrow \quad B = 4$

Similarly put x = –1 on both side

$\Rightarrow \quad C = \frac{1}{2}$

Now that we have found the values of A, B and C, the integration be comes

$\int \frac{-7}{2(x-1)}dx + \int \frac{4dx}{(x-2)} + \int \frac{dx}{2(x+1)}$

$= -\frac{7}{2}\ell n|x-1| + 4\ell n|x-2| + \frac{\ell n|x+1|}{2} + c$

Form	Form of Partial Fraction
$\frac{px+q}{(x-a)(x-b)}$	$\frac{A}{x-a}+\frac{B}{x-b}$
$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}$

$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{(x-a)}+\frac{B}{(x-b)}+\frac{C}{(x-c)}$
$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$ where x^2+bx+c can't be factorized	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+C}$

For cases, where denominate is repeated linear factor of x, we can solve it like given below:

Ex. $\frac{x^2+2x+4}{(x-1)(x+1)^2}=\frac{A}{x-1}+\frac{B}{x+1}+\frac{C}{(x+1)^2}$

Now compare the coefficient

$x^2+2x+4=A(x+1)^2+B(x^2-1)+C(x-1)$

$x=-1\Rightarrow C=-\frac{3}{2}$

$x=1\Rightarrow A=\frac{7}{4}$

$x=0\Rightarrow 4=A-B-C$

$\Rightarrow B=-\frac{3}{4}$

Now we can easily integrate it.

Similarly, $\frac{x^2+2x+4}{(x-1)(x+1)^3}$, can be written as

$\frac{A}{x-1}+\frac{B}{x+1}+\frac{C}{(x+1)^2}+\frac{D}{(x+1)^3}$

For cases, where denominator is quadratic in 'x' which cannot be factorized:

Ex: $\dfrac{x^2+2x+4}{(x-1)(x^2+x+1)}=\dfrac{A}{x-1}+\dfrac{Bx+C}{x^2+x+1}$

$\Rightarrow x^2+2x+4=A(x^2+x+1)+(Bx+C)(x-1)$

Put $x=1\Rightarrow A=\dfrac{7}{3}$

Put $x=0\Rightarrow A-C=4\Rightarrow C=\dfrac{-5}{3}$

Comparing coefficient of $x^2, 1=A+B\Rightarrow B=-\dfrac{4}{3}$

$$\Rightarrow \int\frac{7}{3(x-1)}dx-\frac{1}{3}\int\frac{4x+5}{x^2+x+1}dx$$

$$=\frac{7}{3}\ell n(x-1)-\frac{1}{3}\int\frac{2(2x+1)}{(x^2+x+1)}dx+\left(\frac{-1}{3}\right)\int\frac{3dx}{x^2+x+1}$$

$$=\frac{7\ell n(x-1)}{3}-\frac{2}{3}\ell n(x^2+x+1)-\int\frac{dx}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{7\ell n(x-1)-2\ell n(x^2+x+1)}{3}-\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+c$$

Ex. $\int\dfrac{x^2+2}{(x+1)(x^2-1)}dx$

Sol. We have $\dfrac{x^2+2}{(x+1)(x^2-1)}=\dfrac{x^2+2}{(x+1)^2(x-1)}=\dfrac{A}{x+1}+\dfrac{B}{(x+1)^2}+\dfrac{C}{x-1}$

i.e. $x^2+2=A(x^2-1)+B(x-1)+C(x+1)^2$

Comparing the coefficient, we have

$A+C=1, B+2C=0$ and $-A-B+C=2$

gives $A=\dfrac{1}{4}, B=\dfrac{-3}{2}$ and $C=\dfrac{3}{4}$

Hence, we have

$$I=\int\frac{x^2+2}{(x+1)(x^2-1)}dx=A\int\frac{dx}{x+1}+B\int\frac{dx}{(x+1)^2}+C\int\frac{dx}{x-1}$$

$$=\frac{1}{4}\ln|x+1|+\frac{3}{4}\ln|x-1|+\frac{3}{2(x+1)}+c$$

Q. $\int \frac{(1+x)^3}{(1-x)^3}dx$

Sol. We have

$$\frac{(1+x)^3}{(1-x)^3} = \frac{x^3+3x^2+3x+1}{-x^3+3x^2-3x+1} + 1 - 1 = \frac{6x^2+2}{-x^3+3x^2-3x+1} - 1 = -1 - \frac{6x^2+2}{(x-1)^3}$$

Note :

Before decomposing into partial fractions, we must ensure that the degree of the numerator is less than the degree of the denominator. Take special note of the method of performing division. Adding 1 to the given fraction cancels out the x^3 them in the numerator, therefore reducing the degree of the numerator.

Now we have $\frac{6x^2+2}{(x-1)^3} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$

i.e. $6x^2+2 = A(x-1)^2 + B(x-1) + C$

Comparing the coefficient, we have

$A = 6, -2A + B = 0$ and $A - B + C = 2$

gives $A = 6, B = 12$ and $C = 8$

Hence, we have

$$I = \int \frac{(1+x)^3}{(1-x)^3}dx = -1\int dx - A\int \frac{dx}{x-1} - B\int \frac{dx}{(x-1)^2} - C\int \frac{dx}{(x-1)^3}$$

$$= -x - 6\ln|x-1| + \frac{12}{x-1} + \frac{4}{(x-1)^2} + D.$$

Ex. $\int \frac{1}{(x+1)(x^2+1)^2}dx$

Sol. We have $\frac{1}{(x+1)(x^2+1)^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$

i.e. $1 = A(x^2+1)^2 + (Bx+C)(x^3+x^2+x+1) + (Dx+E)(x+1)$

Comparing the coefficient, we have

$A+B=0, B+C=0, 2A+B+C+D=0$

$B+C+D+E=0$ and $A+C+E=1$

Gives $A=\frac{1}{4}, B=\frac{-1}{4}, C=\frac{1}{4}, D=\frac{-1}{2}$ and $E=\frac{1}{2}$

Hence, we have

$$I=\int\frac{1}{(x+1)(x^2+1)^2}dx=\frac{1}{4}\int\frac{dx}{x+1}-\frac{1}{4}\int\frac{x-1}{x^2+1}dx-\frac{1}{2}\int\frac{x-1}{(x^2+1)^2}dx$$

$$=\frac{1}{4}\ln|x+1|-\frac{1}{8}\int\frac{2x}{x^2+1}dx+\frac{1}{4}\int\frac{dx}{x^2+1}dx-\frac{1}{4}\int\frac{2x}{(x^2+1)^2}dx+\frac{1}{2}\int\frac{dx}{(x^2+1)^2}$$

$$=\frac{1}{4}\ln|x+1|-\frac{1}{8}\ln(x^2+1)+\frac{1}{4}\tan^{-1}x+\frac{1}{4(x^2+1)}+\frac{1}{2}I_1$$

To evaluate I_1, put $x=\tan\theta$ and $dx=\sec^2\theta d\theta$. Thus, we have

$$I_1=\int\frac{\sec^2\theta d\theta}{\sec^4\theta}=\int\frac{1+\cos 2\theta}{2}d\theta=\frac{1}{2}\left[\theta+\frac{\sin 2\theta}{2}\right]=\frac{1}{2}\left[\tan^{-1}x+\frac{x}{x^2+1}\right]$$

Hence, we have

$$I=\frac{1}{4}\ln|x+1|-\frac{1}{8}\ln(x^2+1)+\frac{1}{2}\tan^{-1}x+\frac{x+1}{4(x^2+1)}+c$$

Note :

$$\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\ell n\left|\frac{a+x}{a-x}\right|$$

$$\int\frac{dx}{x^2-a^2}=\frac{1}{2a}\ell n\left|\frac{x-a}{x+a}\right|$$

Q1 **Solve the following questions :**

(i) $\int\frac{x^3dx}{x^4+3x^2+2}$

Sol. Let $x^2=t$

$$\Rightarrow\ \frac{1}{2}\int\frac{tdt}{t^2+3t+2}$$

$$=\frac{1}{2}\int\left(\frac{A}{t+1}+\frac{B}{t+2}\right)dt$$

$\Rightarrow$ A = –1 and B = 2 (using coefficient comparison)

$$\Rightarrow \frac{1}{2}\int\frac{-1}{t+1}dt+\frac{1}{2}\int\frac{2dt}{t+2}$$

$$=\frac{-1}{2}\ell n\left|t+1\right|+\ell n\left|t+2\right|+C$$

$$=\frac{-1}{2}\ell n\left|x^2+1\right|+\ell n\left|x^2+2\right|+C$$

Q. **(ii)** $\int\frac{dx}{x^3+1}$

Sol. $$\int\frac{dx}{(x+1)(x^2-x+1)}=\int\left[\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1}\right]dx$$

$$\Rightarrow 1=A\left(x^2-x+1\right)+(Bx+C)(x+1)$$

Put $x=-1 \Rightarrow A=\frac{1}{3}$

Put $x=0\Rightarrow A+C=1\Rightarrow C=\frac{2}{3}$

Comparing coefficient of x^2 : $0=A+B\Rightarrow B=\frac{-1}{3}$

$$\Rightarrow \frac{1}{3}\int\left[\frac{1}{x+1}+\frac{(-x+2)}{x^2-x+1}\right]dx$$

$$=\frac{\ell n\left|x+1\right|}{3}-\frac{1}{6}\int\frac{2x-4}{x^2-x+1}dx$$

$$=\frac{\ell n\left|x+1\right|}{3}-\frac{1}{6}\int\left(\frac{2x-1}{x^2-x+1}-\frac{3}{x^2-x+1}\right)dx$$

$$=\frac{\ell n\left|x+1\right|}{3}-\frac{1}{6}\ell n\left|x^2-x+1\right|+\frac{1}{2}\int\frac{dx}{\left(x-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{\ell n\left|x+1\right|}{3}-\frac{1}{6}\ell n(x^2-x+1)+\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)+D$$

Q. (iii) $\int \frac{dx}{\sin x(3+2\cos x)}$

Sol. Let $\cos x = y$

$\Rightarrow -\sin x\,dx = dy$

$$\Rightarrow \int \frac{-dy}{(1-y^2)(3+2y)} = \int \frac{dy}{(y^2-1)(2y+3)}$$

$$\text{Let } \frac{1}{(y^2-1)(2y+3)} = \frac{A}{y-1} + \frac{B}{y+1} + \frac{C}{2y+3}$$

$$\Rightarrow A = \frac{1}{10},\ B = \frac{-1}{2},\ C = \frac{4}{5}$$

$$\Rightarrow \int \frac{dy}{10(y-1)} - \frac{1}{2}\int \frac{dy}{y+1} + \frac{4}{5}\int \frac{dy}{2y+3}$$

$$= \frac{\ell n|y-1|}{10} - \frac{\ell n|y+1|}{2} + \frac{2}{5}\ell n(2y+3) + c$$

$$= \frac{\ell n|1-\cos x|}{10} - \frac{\ell n|1+\cos x|}{2} + \frac{2}{5}\ell n(2\cos x+3) + c$$

Q. (iv) $\int \frac{\sin x}{\sin 4x}dx$

Sol. $$\int \frac{\sin x\,dx}{2\sin 2x\cos 2x} = \int \frac{dx}{4\cos x\cos 2x} = \int \frac{\cos x dx}{4(1-\sin^2 x)(1-2\sin^2 x)}$$

Let $\sin x = y$

$$\Rightarrow \frac{1}{4}\int \frac{dy}{(1-y^2)(1-2y^2)} = \frac{1}{8}\int \frac{dy}{(y^2-1)\left(y^2-\frac{1}{2}\right)}$$

$$= \frac{1}{4}\int \left(\frac{1}{y^2-1} - \frac{1}{y^2-\frac{1}{2}}\right)dy$$

$$= \frac{1}{4}\left(\frac{1}{2}\ell n\left|\frac{\sin x-1}{\sin x+1}\right| - \frac{1}{\sqrt{2}}\ell n\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right|\right) + c$$

Q. (v) $\int \frac{1+x\cos x}{x\left(1-x^2e^{2\sin x}\right)}dx$

Sol. $\int \frac{e^{\sin x}\left(1+x\cos x\right)}{xe^{\sin x}\left(1-x^2e^{2\sin x}\right)}dx$

Let $xe^{\sin x}$ be t

$\Rightarrow \int \frac{dt}{t\left(1-t^2\right)} = \int\left(\frac{A}{t}+\frac{B}{1-t}+\frac{C}{1+t}\right)dt$

$\Rightarrow A = 1, B = \frac{1}{2}, C = \frac{-1}{2}$

$\Rightarrow \int\left[\frac{1}{t}+\frac{1}{2(1-t)}-\frac{1}{2(1+t)}\right]dt$

$= \ell n|t| - \frac{1}{2}\ell n|1-t| - \frac{1}{2}\ell n|1+t| + c$

$= \ell n|xe^{\sin x}| - \frac{1}{2}\ell n|1-x^2e^{2\sin x}| + c$

Q. (vi) $\int \frac{(x-1)(x-2)(x-3)}{(x-4)(x-5)(x-6)}dx$

Sol. It is of form $\frac{\text{cubic}}{\text{cubic}}$

Since $\frac{\text{leading coefficient in numerator}}{\text{leading coefficient in denominator}} = 1$

We can write the above integral as

$\int\left(1+\frac{A}{x-4}+\frac{B}{x-5}+\frac{C}{x-6}\right)dx$

$\Rightarrow (x-1)(x-2)(x-3) = (x-4)(x-5)(x-6) + A(x-5)(x-6) + B(x-4)(x-6) + C(x-4)(x-5)$

$\Rightarrow$ put $x = 4 \Rightarrow 6 = 2A \Rightarrow A = 3$

put $x = 5 \Rightarrow 24 = -B \Rightarrow B = -24$

put $x = 6 \Rightarrow 60 = 2C \Rightarrow C = 30$

$\Rightarrow \int\left(1+\frac{3}{x-4}-\frac{24}{x-5}+\frac{30}{x-6}\right)dx$

$= x + 3\ell n|x-4| - 24\ell n|x-5| + 30\ell n|x-6| + c$

Q2 **Suppose f(x) is a quadratic function such that f(0) = 1 and f(–1) = 4. If $\int \frac{f(x)\,dx}{x^2(x+1)^2}$ is a rational function, find the value of f(10).**

Sol. $\int \frac{f(x)\,dx}{x^2(x+1)^2} = \int\left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2}\right)dx$

But since, integral is a rational function, A = C = 0

as $\int \frac{1}{x}dx$ and $\int \frac{1}{x+1}dx$ is a logarithmic function.

$\Rightarrow \int \frac{f(x)}{x^2(x+1)^2}dx = \int\left(\frac{B}{x^2} + \frac{D}{(x+1)^2}\right)dx$

$\Rightarrow f(x) = B(x+1)^2 + Dx^2$

$f(0) = 1 \Rightarrow B = 1$

$f(-1) = 4 \Rightarrow D = 4$

$\therefore f(x) = (x+1)^2 + 4x^2$

$= 5x^2 + 2x + 1$

$\Rightarrow f(10) = 521$

Point to Remember!!!

Trick:
Idea is to change the integrand such that the substitution becomes visible.

SUBSTITUTION AFTER MANIPULATION (KUTURPUTUR)

Ex. $\int \frac{dx}{x(x^2+1)}$

Sol. $\int \frac{dx}{x^3\left(1+\frac{1}{x^2}\right)}$

Let $1 + \frac{1}{x^2} = t$

$\Rightarrow \frac{-2dx}{x^3} = dt$

$\Rightarrow \frac{-1}{2}\int \frac{dt}{t} = \frac{-\ell n|t|}{2} + C = \frac{-\ell n\left|\frac{x^2+1}{x^2}\right|}{2} + C$

Ex. $\int \frac{dx}{x\left(x^n+1\right)}$

Sol. $\int \frac{dx}{x\left(x^n+1\right)} = \int \frac{dx}{x^{n+1}\left(1+\frac{1}{x^n}\right)}$

Put $1+\frac{1}{x^n} = t$ to get $\int \frac{-\frac{1}{n}dt}{t} = -\frac{1}{n}\ln t + c = -\frac{1}{n}\ln\left(1+\frac{1}{x^n}\right)+c$

Ex. $\int \frac{x^7}{\left(1-x^2\right)^5}dx$

Sol. $\int \frac{x^7 dx}{x^{10}\left(\frac{1}{x^2}-1\right)^5} = \int \frac{dx}{x^3\left(\frac{1}{x^2}-1\right)^5}$

Put $\frac{1}{x^2}-1 = t$ to get $\int \frac{-\frac{1}{2}}{t^5}dt = \left(\frac{-1}{2}\right)\left(\frac{-1}{4}\right)t^{-4}+c = \frac{1}{8}\frac{1}{\left(x^{-2}-1\right)^4}+c$

Ex. $\int \frac{xdx}{\left(1-x^4\right)^{\frac{3}{2}}}$

Sol. $I = \int \frac{xdx}{x^6\left(\frac{1}{x^4}-1\right)^{\frac{3}{2}}} = \int \frac{dx}{x^5\left(\frac{1}{x^4}-1\right)^{\frac{3}{2}}}$

Put $\frac{1}{x^4}-1 = t \qquad \Rightarrow -4\frac{1}{x^5}dx = dt$

$\Rightarrow \quad I = \int \frac{-\frac{1}{4}dt}{t^{\frac{3}{2}}} = \frac{1}{2}\frac{1}{\sqrt{t}}+c = \frac{1}{2}\frac{1}{\sqrt{\frac{1}{x^4}-1}}$

Q. **Solve the following :**

(i) $\int \frac{dx}{x^4\left(x^3+1\right)^2}$

Sol. $\int \frac{dx}{x^4.x^6\left(1+x^{-3}\right)^2}$

Let $1+x^{-3}=t$

$\Rightarrow \frac{-1}{3}\int \frac{\left(t-1\right)^2 dt}{t^2}$

$\Rightarrow -\frac{1}{3}\int\left(1-\frac{2}{t}+\frac{1}{t^2}\right)dt$

$= -\frac{t}{3}+\frac{2\ell n\left|t\right|}{3}+\frac{1}{3t}+C$

$= -\frac{\left(1+x^{-3}\right)}{3}+\frac{2\ell n\left|1+x^{-3}\right|}{3}+\frac{1}{3\left(1+x^{-3}\right)}+C$

Q. **(ii)** $\int \frac{dx}{x^2\left(x+\sqrt{1+x^2}\right)}$

Sol. $\int \frac{dx}{x^2.x\left(1+\sqrt{1+\frac{1}{x^2}}\right)}$

Let $1+\frac{1}{x^2}=y^2$

$\Rightarrow \frac{-2}{x^3}dx = 2ydy$

$\Rightarrow \int \frac{-ydy}{1+y}$

$= -\int\left(1-\frac{1}{1+y}\right)dy$

$= \ell n\left|1+y\right|-y+C$

$= \ell n\left(1+\sqrt{1+\frac{1}{x^2}}\right)-\sqrt{1+\frac{1}{x^2}}+C$

Q. (iii) $\int \frac{\left(ax^2 - b\right)dx}{x\sqrt{c^2x^2 - \left(ax^2 + b\right)^2}}$

Sol. $\int \frac{\left(ax - \frac{b}{x}\right)dx}{\sqrt{c^2x^2 - x^2\left(ax + \frac{b}{x}\right)^2}}$

$= \int \frac{\left(a - \frac{b}{x^2}\right)dx}{\sqrt{c^2 - \left(ax + \frac{b}{x}\right)^2}}$

Let $ax + \frac{b}{x} = y$

$\Rightarrow \int \frac{dy}{\sqrt{c^2 - y^2}} = \sin^{-1}\left(\frac{y}{c}\right) + k$

$= \sin^{-1}\left(\frac{ax^2 + b}{cx}\right) + k$

Q. (iv) $\int \frac{x\cos\alpha + 1}{\left(x^2 + 2x\cos\alpha + 1\right)^{\frac{3}{2}}}$

Sol. $\int \frac{x^3\left(\frac{\cos\alpha}{x^2} + \frac{1}{x^3}\right)}{x^3\left(1 + \frac{2\cos\alpha}{x} + \frac{1}{x^2}\right)^{\frac{3}{2}}}dx$

Let $1 + \frac{2\cos\alpha}{x} + \frac{1}{x^2} = y$

$\Rightarrow -2\left(\frac{\cos\alpha}{x^2} + \frac{1}{x^3}\right)dx = dy$

$\Rightarrow \int -\frac{1}{2}\frac{dy}{y^{\frac{3}{2}}} = \frac{1}{\sqrt{y}} + c$

$= \frac{1}{\sqrt{1 + \frac{2\cos\alpha}{x} + \frac{1}{x^2}}} + c$

Q. **(v)** $\int \frac{(x-1)dx}{x^2\sqrt{2x^2-2x+1}}$

Sol.

$$\int \frac{\left(\frac{1}{x^2}-\frac{1}{x^3}\right)dx}{\sqrt{2-\frac{2}{x}+\frac{1}{x^2}}}$$

let $2-\frac{2}{x}+\frac{1}{x^2}=y^2$

$\Rightarrow \left(\frac{1}{x^2}-\frac{1}{x^3}\right)dx = ydy$

$\Rightarrow \int \frac{ydy}{y} = y+c$

$= \sqrt{2-\frac{2}{x}+\frac{1}{x^2}}+C$

Q. **(vi)** $\int \left(x^6+x^4+x^2\right)\left(2x^4+3x^2+6\right)^{\frac{1}{2}}dx$

Sol. Degree in both expression are as follows

(6 4 2) (4 2 0)

↓ If one take x from 1st and put it in 2nd

(5 3 1) (6 4 2)

Now above form is ideal for kuturputur

$\Rightarrow \int \left(x^5+x^3+x\right)\left(2x^6+3x^4+6x^2\right)^{\frac{1}{2}}dx$

Let $2x^6+3x^4+6x^2=y^2$

$\Rightarrow 12\left(x^5+x^3+x\right)dx = 2ydy$

$\Rightarrow \int \frac{y^2}{6}dy = \frac{y^3}{18}+C$

$= \frac{\left(2x^6+3x^4+6x^2\right)^{\frac{3}{2}}}{18}+C$

Q. **(vii)** $\int \frac{2+\sqrt{x}}{\left(x+1+\sqrt{x}\right)^2}dx$

Sol. $\int \frac{\left(2+\sqrt{x}\right)dx}{x^2\left(1+\frac{1}{x}+\frac{1}{\sqrt{x}}\right)^2}$

$$= \int \frac{\left(\frac{2}{x^2}+\frac{1}{x^{\frac{3}{2}}}\right)dx}{\left(1+\frac{1}{x}+\frac{1}{\sqrt{x}}\right)^2}$$

Let $1+\frac{1}{x}+\frac{1}{\sqrt{x}} = y$

$$\Rightarrow \left(\frac{-1}{x^2}+\frac{-1}{2x^{\frac{3}{2}}}\right)dx = dy$$

$$\Rightarrow \int \frac{-2dy}{y^2} = \frac{2}{y} + C$$

$$= \frac{2}{\left(1+\frac{1}{x}+\frac{1}{\sqrt{x}}\right)^2} + C$$

Q. **(viii)** $\int \frac{1}{\left(1+\sqrt{x}\right)^8} dx$

Sol. $\int \frac{\sqrt{x}\ dx}{\left(1+\sqrt{x}\right)^8 \sqrt{x}}$

Let $1+\sqrt{x} = y$

$$\Rightarrow \frac{dx}{2\sqrt{x}} = dy$$

$$\Rightarrow \int \frac{2(y-1)}{y^8} dy = \int 2\left(\frac{1}{y^7}-\frac{1}{y^8}\right)dy$$

$$= \frac{-1}{3y^6} + \frac{2}{7y^7} + C$$

$$= \frac{-1}{3\left(1+\sqrt{x}\right)^6} + \frac{2}{7\left(1+\sqrt{x}\right)^7} + C$$

Q. **(ix)** $\int \frac{x^4 + x^8}{\left(1 - x^4\right)^{\frac{7}{2}}} dx$

Sol. $\int \frac{x^7\left(x^{-3} + x\right)}{x^7\left(x^{-2} - x^2\right)^{7/2}} dx$

Let $\frac{1}{x^2} - x^2 = y$

$\Rightarrow \quad -2\left(x^{-3} + x\right)dx = dy$

$\Rightarrow \quad \int \frac{-dy}{2y^{\frac{7}{2}}} = \frac{y^{\frac{-5}{2}}}{5} + C = \frac{1}{5\left(\frac{1}{x^2} - x^2\right)^{\frac{5}{2}}} + C$

Q. **(x)** $\int \frac{1}{\left(1 + x^4\right)\left(\sqrt{\sqrt{1 + x^4} - x^2}\right)} dx$

Sol. $\int \frac{dx}{x^4\left(x^{-4} + 1\right)x\sqrt{\sqrt{x^{-4} + 1} - 1}}$

Let $x^{-4} + 1 = y^2$

$\Rightarrow \quad \frac{-4}{x^5} dx = 2ydy$

$\Rightarrow \quad \frac{-1}{2}\int \frac{ydy}{y^2\sqrt{y - 1}}$

$= \frac{-1}{2}\int \frac{dy}{y\sqrt{y - 1}}$

let $y = \sec^2\theta$

$\Rightarrow \quad \frac{-1}{2}\int \frac{2\sec^2\theta \tan\theta d\theta}{\sec^2\theta \tan\theta}$

$= -\int d\theta = -\theta + C = -\sec^{-1}\sqrt{y} + C$

$= -\sec^{-1}\sqrt{\sqrt{1 + x^{-4}}} + C$

Q. **(xi)** $\int \frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}}$

Sol. Let $x-1=y$

$$\Rightarrow \int \frac{dy}{\left(y^3(y+3)^5\right)^{\frac{1}{4}}}$$

$$\Rightarrow \int \frac{1}{y^2} \frac{dy}{\left(1+\frac{3}{y}\right)^{\frac{5}{4}}}$$

Let $1+\frac{3}{y}=t$

$$\Rightarrow -\frac{1}{3}\int \frac{dt}{t^{\frac{5}{4}}}$$

$$= \frac{4}{3}\left(1+\frac{3}{y}\right)^{-\frac{1}{4}}+C \qquad = \frac{4}{3}\left(\frac{x+2}{x-1}\right)^{-\frac{1}{4}}+C \qquad = \frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}}+C$$

Q. **(xii)** $\int \frac{x^2}{(x\sin x+\cos x)^2}dx$

Sol. $\frac{d}{dx}(x\sin x+\cos x)=x\cos x$

$$\Rightarrow \int \frac{x\cos x}{(x\sin x+\cos x)^2}\times(x\sec x)dx$$

Now applying by parts,

$$\Rightarrow = \frac{-x\sec x}{x\sin x+\cos x}+\int \frac{\sec x+x\sec x\tan x}{x\sin x+\cos x}dx$$

$$= \frac{-x\sec x}{x\sin x+\cos x}+\int \frac{\sec x(x\sin x+\cos x)}{\cos x(x\sin x+\cos x)}dx$$

$$= \frac{-x\sec x}{x\sin x+\cos x}+\tan x+C$$

INTEGRALS OF TRIGONOMETRIC FUNCTION

Type-1:

$$\int \frac{dx}{a+b\sin^2 x} \Big/ \int \frac{dx}{a+b\cos^2 x} \Big/ \int \frac{dx}{a\sin^2 x+b\cos^2 x+c\sin x\cos x} \Big/ \int \frac{dx}{(a\cos x+b\sin x)^2}$$

Multiply N^r and D^r by $\sec^2 x$ or $\text{cosec}^2 x$ and proceed

Type-2:

$$\int \frac{dx}{a+b\sin x} \Big/ \int \frac{dx}{a+b\cos x} \Big/ \int \frac{dx}{a\sin x+b\cos x+c}$$

Convert $\sin x$ and $\cos x$ into their corresponding tangent to half the angles and put $\tan\frac{x}{2}=t$

Type-3

$$\int \frac{a\sin x+b\cos x+c}{\ell\sin x+m\cos x+n}dx;\quad N^r = A\left(D^r\right)+B\left(\frac{d}{dx}D^r\right)+C$$

Type-4

$$\int \frac{x^2+1}{x^4+kx^2+1}dx \text{ or } \int \frac{x^2-1}{x^4+kx^2+1}dx$$

Divide N^r and D^r by x^2 and take suitable substitution

Ex. $\int \frac{dx}{4-5\sin^2 x}$

Sol. $\int \frac{dx}{4-5\sin^2 x} = \int \frac{\text{cosec}^2 x}{4\cot^2 x-1}dx$

Put $\cot x = t$ to get

$$\int \frac{-dt}{4t^2-1} = \frac{-1}{4}\int \frac{dt}{t^2-\left(\frac{1}{2}\right)^2} = \frac{-1}{4}\ln\left(\frac{t-\frac{1}{2}}{t+\frac{1}{2}}\right)+c = -\frac{1}{4}\ln\left(\frac{2\cot x-1}{2\cot x+1}\right)+c$$

Ex. $\int \frac{dx}{5+4\cos x}$

Sol. $\int \frac{dx}{5+4\cos x} = \int \frac{dx}{5+4\frac{\left(1-\tan^2\frac{x}{2}\right)}{\left(1+\tan^2\frac{x}{2}\right)}} = \int \frac{\sec^2\frac{x}{2}}{9+\tan^2\frac{x}{2}}dx$

Put $\tan\frac{x}{2}=t$ to get $\int\frac{2dt}{9+t^2}=\frac{2}{3}\tan^{-1}\left(\frac{t}{3}\right)+c$

Ex. $\int\frac{1}{\sin x-3\cos x-1}dx$

Sol. Let $I=\int\frac{1}{\sin x-3\cos x-1}dx$

Putting $\tan\frac{x}{2}=t$, we have

$$I=\int\frac{1}{\frac{2t}{1+t^2}-3\left(\frac{1-t^2}{1+t^2}\right)-1}\cdot\frac{2dt}{1+t^2}=\int\frac{2dt}{2t-3\left(1-t^2\right)-\left(1+t^2\right)}$$

$$=\int\frac{2dt}{2t^2+2t-4}=\int\frac{dt}{(t-1)(t+2)}=\frac{1}{3}\int\left[\frac{1}{t-1}-\frac{1}{t+2}\right]dt=\frac{1}{3}\ln\left|\frac{t-1}{t+2}\right|+C$$

$$=\frac{1}{3}\ln\left|\frac{\tan\frac{x}{2}-1}{\tan\frac{x}{2}+2}\right|+C$$

Ex. **Evaluate the indefinite integral** $\int\frac{\sin x+3\cos x+1}{\sin x-3\cos x-1}dx$

Sol. Let $\sin x+3\cos x+1=\lambda\left(\sin x-3\cos x-1\right)+\mu\left(\cos x+3\sin x\right)+v$

Comparing the coefficient of sin x, cos x and constant term we have

$\lambda+3\mu=1,-3\lambda+\mu=3$ and $-\lambda+v=1$, it gives $\lambda=-\frac{4}{5}$ and $v=\frac{1}{5}$ and $\mu=\frac{3}{5}$

Thus, we have

$$I=\frac{-4}{5}\int 1dx+\frac{3}{5}\int\frac{\cos x+3\sin x}{\sin x-3\cos x-1}dx+\frac{1}{5}\int\frac{dx}{\sin x-3\cos x-1}$$

$$=\frac{-4}{5}x+\frac{3}{5}\ln\left|\sin x-3\cos x-1\right|+\frac{1}{5}I_1$$

Now, we have

$$I_1=\int\frac{1}{\frac{2t}{1+t^2}-\frac{3\left(1-t^2\right)}{1+t^2}-1}\cdot\frac{2dt}{1+t^2}\qquad\left[\text{Putting }\tan\frac{x}{2}=t\right]$$

$$\int\frac{2dt}{2t-3\left(1-t^2\right)-\left(1+t^2\right)}=\int\frac{dt}{2t^2+2t-4}=\int\frac{dt}{(t-1)(t+2)}=\frac{1}{3}\int\left(\frac{1}{t-1}-\frac{1}{t+2}\right)dt$$

$$= \frac{1}{3}\ln\left|\frac{t-1}{t+2}\right| = \frac{1}{3}\ln\left|\frac{\tan\left(\frac{x}{2}\right)-1}{\tan\left(\frac{x}{2}\right)+2}\right|$$

Hence, we have

$$I = \frac{-4}{5}x + \frac{3}{5}\ln|\sin x - 3\cos x - 1| + \frac{1}{15}\ln\left|\frac{\tan\left(\frac{x}{2}\right)-1}{\tan\left(\frac{x}{2}\right)+2}\right| + C$$

Ex. $\int \frac{dx}{(3\sin x - 4\cos x)^2}$

Sol. $I = \int \frac{dx}{(3\sin x - 4\cos x)^2} = \int \frac{\sec^2 x dx}{(3\tan x - 4)^2}$

Put $3\tan x - 4 = t \Rightarrow 3\sec^2 x\, dx = dt$

$$\Rightarrow \quad I = \int \frac{\frac{1}{3}dt}{t^2} = -\frac{1}{3}\frac{1}{(3\tan x - 4)} + c$$

Q. $\int \frac{x^2+1}{x^4+7x^2+1}dx$

Sol. $\int \frac{x^2+1}{x^4+7x^2+1}dx = \int \frac{\left(1+\frac{1}{x^2}\right)}{x^2+7+\frac{1}{x^2}}dx$

Put $x - \frac{1}{x} = t \Rightarrow \left(1+\frac{1}{x^2}\right)dx = dt$ & $x^2 + \frac{1}{x^2} = t^2 + 2$

To get $\int \frac{dt}{t^2+9} = \frac{1}{3}\tan^{-1}\left(\frac{t}{3}\right) + c = \frac{1}{3}\tan^{-1}\left(\frac{x^2-1}{3x}\right) + c$

Q. $\int \frac{x^2+1}{x^4+x^2+1}dx$

Sol. $I = \int \frac{x^2+1}{x^4+x^2+1}dx = \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}+1}dx = \int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+3}dx$

$$= \int \frac{dt}{t^2+3} \qquad \left[\text{Putting } x - \frac{1}{x} = t \text{ and } \left(1+\frac{1}{x^2}\right)dx = dt\right]$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{t}{\sqrt{3}}\right) + C = \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x^2-1}{\sqrt{3}x}\right) + C$$

Ex. $\int \frac{x^2+2}{x^4-5x^2+4} dx$

Sol. $I = \int \frac{x^2+2}{x^4-5x^2+4} dx = \int \frac{1+\frac{2}{x^2}}{x^2+\frac{4}{x^2}-5} dx = \int \frac{1+\frac{2}{x^2}}{\left(x-\frac{2}{x}\right)^2-1} dx$

$$= \int \frac{dt}{t^2-1} \qquad \left[\text{Putting } x - \frac{2}{x} = t \text{ and } \left(1+\frac{2}{x^2}\right)dx = dt\right]$$

$$= \frac{1}{2} \ln\left|\frac{t-1}{t+1}\right| + C = \frac{1}{2} \ln\left|\frac{x^2-x-2}{x^2+x-2}\right| + C$$

Q. **Solve the following :** **(Type-I)**

(i) $\int \frac{dx}{(3\sin x - 4\cos x)^2}$

Sol. $\int \frac{\sec^2 x\, dx}{(3\tan x - 4)^2}$ (∵ Multiplying by $\sec^2 x$)

$$= \frac{1}{3}\left(\frac{1}{4-3\tan x}\right) + C$$

Q. **(ii)** $\int \frac{dx}{3+\cos^2 x}$

Sol. $\int \frac{\sec^2 x dx}{3\sec^2 x + 1}$

$$= \int \frac{\sec^2 x dx}{4+3\tan^2 x}$$

Let $\tan x = t$

$$\Rightarrow \int \frac{dt}{3t^2+4} = \frac{1}{3}\int \frac{dt}{t^2+\left(\frac{2}{\sqrt{3}}\right)^2}$$

$$= \frac{1}{3}\left(\frac{\sqrt{3}}{2}\right)\tan^{-1}\left(\frac{\sqrt{3}t}{2}\right)+C$$

$$= \frac{\sqrt{3}}{6}\tan^{-1}\left(\frac{\sqrt{3}t}{2}\right)+C$$

Q. **Solve the following :** **(Type-II)**

(i) $\int \frac{dx}{4+5\cos x}$

Sol.

$$\int \frac{dx}{4+5\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}$$

$$= \int \frac{\sec^2\left(\frac{x}{2}\right)dx}{9-\tan^2\frac{x}{2}}$$

Let $\tan\frac{x}{2} = t$

$$\Rightarrow \int \frac{2dt}{9-t^2} = -2\int \frac{dt}{t^2-3^2}$$

$$= \frac{-2}{2\times 3}\ln\left|\frac{t-3}{t+3}\right|+C$$

$$= \frac{1}{3}\ell n\left|\frac{\tan\frac{x}{2}+3}{\tan\frac{x}{2}-3}\right|+C$$

Q. **(ii)** $\int \frac{dx}{3+2\sin x+\cos x}$

Sol.

$$\int \frac{dx}{3+2\left(\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)+\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}$$

$$= \int \frac{\sec^2\left(\frac{x}{2}\right)}{2\tan^2\left(\frac{x}{2}\right)+4\tan\frac{x}{2}+4}dx$$

Let $\tan\frac{x}{2} = t$

$$\Rightarrow \int \frac{dt}{t^2+2t+2} = \int \frac{dt}{(t+1)^2+1}$$

$$= \tan^{-1}(t+1) + C$$

$$= \tan^{-1}\left(\tan\frac{x}{2}+1\right) + C$$

Q. **(iii)** $\int \frac{dx}{\cos x(5+3\cos x)}$

Sol.

$$\frac{3}{5}\int \frac{(3\cos x+5)-(3\cos x)}{(3\cos x)(3\cos x+5)}dx$$

$$= \frac{1}{5}\int \frac{dx}{\cos x} - \frac{3}{5}\int \frac{dx}{3\left(\frac{1-\tan^2\left(\frac{x}{2}\right)}{1+\tan^2\left(\frac{x}{2}\right)}\right)+5}$$

$$= \frac{\ell n|\sec x+\tan x|}{5} - \frac{3}{5}\int \frac{\sec^2\frac{x}{2}dx}{2\tan^2\frac{x}{2}+8}$$

Let $\tan\frac{x}{2}$ be t

$$\Rightarrow \frac{\ell n|\sec x+\tan x|}{5} - \frac{3}{5}\int \frac{dt}{t^2+4}$$

$$= \frac{\ell n|\sec x+\tan x|}{5} - \frac{3}{10}\tan^{-1}\left(\frac{\tan\frac{x}{2}}{2}\right) + C$$

Q. **(iv)** $\int \frac{dx}{1-\cos x+\sin x}$

Sol.

$$\int \frac{dx}{1-\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)+\left(\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}$$

$$= \int \frac{\sec^2\left(\frac{x}{2}\right)dx}{2\tan^2\left(\frac{x}{2}\right)+2\tan\frac{x}{2}}$$

$$= \int \frac{dt}{t(t+1)} \qquad \text{where } t = \tan\frac{x}{2}$$

$$= \int\left(\frac{1}{t}-\frac{1}{t+1}\right)dt$$

$$= \ell n\left(\frac{t}{t+1}\right)+C$$

$$= \ell n\left(\frac{\tan\frac{x}{2}}{\tan\frac{x}{2}+1}\right)+C$$

Q. **(v)** $\int \frac{\sin 2x}{\sin^4 x+\cos^4 x}dx$

Sol. $\int \frac{\sin 2x\ dx}{1-\frac{1}{2}(\sin 2x)^2}$

$$= \int \frac{\sin 2x}{\frac{1}{2}+\frac{(\cos 2x)^2}{2}}dx$$

Let cos2x=t

$$\Rightarrow \int \frac{-dt}{1+t^2} = -\tan^{-1}(t)+C$$

$$= -\tan^{-1}(\cos 2x)+C$$

Q. **(vi)** $\int \frac{dx}{\sin^4 x+\cos^4 x}$

Sol. $\int \frac{dx}{1-\frac{1}{2}\sin^2 2x}$

$$= \int \frac{\sec^2 2x\ dx}{\sec^2 2x-\frac{\tan^2 2x}{2}}$$

$$= \int \frac{\sec^2 2x \ dx}{1 + \frac{\tan^2 2x}{2}}$$

Let $\tan 2x = t$

$$\Rightarrow \int \frac{dt}{t^2 + 2} = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t}{\sqrt{2}}\right) + C$$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\tan 2x}{\sqrt{2}}\right) + C$$

Q. **Solve the following :** **(Type-III)**

(i) $\int \frac{11\cos x - 16\sin x}{2\cos x + 5\sin x} dx$

Sol. $11\cos x - 16\sin x = A(2\cos x + 5\sin x) + B(-2\sin x + 5\cos x)$

$\Rightarrow$ $2A + 5B = 11$ and $5A - 2B = -16$

Solving, $A = -2$ and $B = 3$

$$\Rightarrow \int \left[-2 + \frac{3(-2\sin x + 5\cos x)}{2\cos x + 5\sin x}\right] dx$$

$$= -2x + 3\ell n\left|2\cos x + 5\sin x\right| + C$$

Q. **(ii)** $\int \frac{1}{1 + \tan x} dx$

Sol. $$\int \frac{\cos x}{\sin x + \cos x} = \frac{1}{2}\int \frac{(\sin x + \cos x) + (\cos x - \sin x)}{\sin x + \cos x} dx$$

$$= \frac{1}{2}\int \left(1 + \frac{\cos x - \sin x}{\sin x + \cos x}\right) dx$$

$$= \frac{x}{2} + \frac{\ell n\left|\sin x + \cos x\right|}{2} + C$$

Q. **(iii)** $\int \frac{1}{1 + \cot x} dx$

Sol. $$\int \frac{\sin x}{\sin x + \cos x} dx$$

$$= \frac{1}{2}\int \left(1 + \frac{\sin x - \cos x}{\sin x + \cos x}\right) dx$$

$$= \frac{x}{2} - \frac{\ell n\left|\sin x + \cos x\right|}{2} + C$$

Q. (iv) $\int \frac{1}{1+\operatorname{cosec} x}\,dx$

Sol. $\int \frac{\sin x}{\sin x+1}\,dx$

$= \int \frac{\sin x(1-\sin x)}{\cos^2 x}\,dx$

$= \int (\sec x \tan x - \tan^2 x)\,dx$

$= \sec x - (\tan x - x) + C$

$= \sec x - \tan x + x + C$

Q. (v) $\int \frac{1}{1+\sec x}\,dx$

Sol. $\int \frac{\cos x}{\cos x+1}\,dx$

$= \int \frac{\cos x(1-\cos x)}{\sin^2 x}\,dx$

$= \int \operatorname{cosec} x \cot x - \cot^2 x\,dx$

$= \int (\operatorname{cosec} x \cot x - \operatorname{cosec}^2 x + 1)\,dx$

$= -\operatorname{cosec} x + \cot x + x + C$

Q. (vi) $\int \frac{6+3\sin x+14\cos x}{3+4\sin x+5\cos x}\,dx$

Sol. $6+3\sin x+14\cos x = A(3+4\sin x+5\cos x)+B(4\cos x-5\sin x)+C$

$\Rightarrow$ $4A-5B=3$

$\Rightarrow$ $5A+4B=14$

$\Rightarrow$ $3A+C=6$

$\Rightarrow$ $A=2, B=1, C=0$

$\Rightarrow$ $2x+\ell n|3+4\sin x+5\cos x|+c$

Q. (vii) $\int \frac{\sin x\,dx}{e^x-\sin x-\cos x}$

Sol. $\frac{d}{dx}(e^x-\sin x-\cos x) = e^x-\cos x+\sin x$

So, if $e^x-\sin x-\cos x = f(x)$

$$\Rightarrow \quad \sin x = \frac{f'(x) - f(x)}{2}$$

$$\Rightarrow \quad \int \frac{f'(x) - f(x)}{2f(x)} dx$$

$$= \frac{\ell n|f(x)|}{2} - \frac{x}{2} + C$$

$$= \frac{\ell n\left(e^x - \sin x - \cos x\right)}{2} - \frac{x}{2} + C$$

Q. **(viii)** $\int \frac{3e^x + 5e^{-x}}{4e^x - 5e^{-x}} dx$

Sol. $3e^x + 5e^{-x} = A\left(4e^x - 5e^{-x}\right) + B\left(4e^x + 5e^{-x}\right)$

$$\Rightarrow \quad 4A + 4B = 3$$
$$-5A + 5B = 5$$

$$\Rightarrow \quad A = \frac{-1}{8} \text{ and } B = \frac{7}{8}$$

$$\Rightarrow \quad \frac{-1}{8}\int dx + \frac{7}{8}\int \frac{4e^x + 5e^{-x}}{4e^x - 5e^{-x}} dx$$

$$= \frac{-x + 7\ell n\left|4e^x - 5e^{-x}\right|}{8} + C$$

Q. **(ix)** $\int \cos 2x . \ell n\left(1 + \tan x\right) dx$

Sol. Applying by-parts,

$$\Rightarrow \quad \frac{\sin 2x}{2} \ell n\left(1 + \tan x\right) - \int \frac{\sin 2x}{2}\left(\frac{\sec^2 x}{1 + \tan x}\right) dx$$

$$\Rightarrow \quad \frac{\sin 2x \ell n\left(1 + \tan x\right)}{2} - \int \frac{\tan x}{1 + \tan x} dx = \frac{\sin 2x\, \ell n\,|1 + \tan x|}{2} - \int \frac{dx}{1 + \cot x}$$

$$= \frac{\sin 2x \ell n\left|1 + \tan x\right|}{2} - \frac{x}{2} + \frac{\ell n\left|\sin x + \cos x\right|}{2} + C \qquad \left(\because \text{previous Q(iii)}\right)$$

Q. **Solve the following :** **(Type-IV)**

(i) $\int \frac{x^2-1}{x^4-23x^2+1}dx$

Sol. $\int \frac{1-\frac{1}{x^2}}{x^2-23+\frac{1}{x^2}}dx$

Let $x+\frac{1}{x}=y$

$$=\int \frac{dy}{y^2-25}=\frac{1}{10}\ell n\left|\frac{y-5}{y+5}\right|+C$$

$$=\frac{1}{10}\ell n\left|\frac{x+\frac{1}{x}-5}{x+\frac{1}{x}+5}\right|+C$$

$$=\frac{1}{10}\ell n\left|\frac{x^2-5x+1}{x^2+5x+1}\right|+C$$

Q. **(ii)** $\int \frac{x^2}{x^4+1}dx$

Sol. $\frac{1}{2}\int \frac{x^2+1}{x^4+1}dx+\frac{1}{2}\int \frac{x^2-1}{x^4+1}dx$

$$=\frac{1}{2}\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx+\frac{1}{2}\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$$

Let $x-\frac{1}{x}=t \quad \& \quad x+\frac{1}{x}=y$

$$\Rightarrow \frac{1}{2}\int \frac{dt}{t^2+2}+\frac{1}{2}\int \frac{dy}{y^2-2}$$

$$=\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)+\frac{1}{4\sqrt{2}}\ell n\left|\frac{y-\sqrt{2}}{y+\sqrt{2}}\right|+C$$

$$=\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)+\frac{1}{4\sqrt{2}}\ell n\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right|+C$$

Q. **(iii)** $\int \sqrt{\tan\theta}\,d\theta$

Sol. Let $\tan\theta = y^2 \Rightarrow \sec^2\theta d\theta = 2ydy$

$$\Rightarrow \int \frac{y.2y\,dy}{1+y^4}$$

$$= \int \frac{2y^2\,dy}{1+y^4}$$

$$= \int \frac{y^2+1dy}{1+y^4} + \int \frac{y^2-1}{1+y^4}dy$$

$$= 2\left(\frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{y-\frac{1}{y}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ell n \left| \frac{y+\frac{1}{y}-\sqrt{2}}{y+\frac{1}{y}+\sqrt{2}} \right| + C \right)$$

Where y is $\sqrt{\tan\theta}$

Q. **(iv)** $\int \sqrt{\cot\theta}\,d\theta$

Sol. Let $\int \sqrt{\tan\theta}\,d\theta = I_1$ and $\int \sqrt{\cot\theta}\,d\theta = I_2$

$$I_1 + I_2 = \int \sqrt{\frac{\sin\theta}{\cos\theta}} + \sqrt{\frac{\cos\theta}{\sin\theta}}d\theta$$

$$= \int \frac{\sin\theta+\cos\theta}{\sqrt{\sin\theta\cos\theta}}d\theta$$

Let $y = \sin\theta - \cos\theta$

$$\Rightarrow I_1 + I_2 = \int \frac{dy}{\sqrt{\frac{1-y^2}{2}}} = \sqrt{2}\sin^{-1} y + C$$

$$\Rightarrow I_1 + I_2 = \sqrt{2}\sin^{-1}(\sin\theta - \cos\theta) + C \qquad ...(1)$$

$$\Rightarrow I_1 - I_2 = \int \left(\sqrt{\frac{\sin\theta}{\cos\theta}} - \sqrt{\frac{\cos\theta}{\sin\theta}} \right) d\theta$$

$$= \int \frac{\sin\theta-\cos\theta}{\sqrt{\sin\theta\cos\theta}}d\theta$$

Let $y = \sin\theta + \cos\theta$

$$\Rightarrow \quad I_1 - I_2 = \int \frac{-dy}{\sqrt{\frac{y^2-1}{2}}} = -\sqrt{2}\ell n\left|y+\sqrt{y^2-1}\right| + C$$

$$\Rightarrow \quad I_1 - I_2 = -\sqrt{2}\ell n\left|\sin\theta+\cos\theta+\sqrt{2\sin\theta\cos\theta}\right| + C \qquad ...(2)$$

From 1 & 2

$$I_2 = \int \sqrt{\cot\theta}d\theta = \frac{\sqrt{2}\sin^{-1}(\sin\theta-\cos\theta)+\sqrt{2}\ell n\left|\sin\theta+\cos\theta+\sqrt{\sin 2\theta}\right|}{2} + C$$

Q. **(v)** $\int \frac{x^{17}}{1+x^{24}}dx$

Sol. Let $x^6 = y$

$$\Rightarrow \int \frac{y^2}{6\left(1+y^4\right)}dy$$

$$= \frac{1}{6}\left(\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{y-\frac{1}{y}}{\sqrt{2}}\right)+\frac{1}{4\sqrt{2}}\ell n\left|\frac{y+\frac{1}{y}-\sqrt{2}}{y+\frac{1}{y}+\sqrt{2}}\right|+C\right) \qquad (\because \text{ Q(ii)})$$

Where $y = x^6$

Note :

(1) $\cos \pm \sin x$ is loving in numerator if denominator contains $'a+\sin 2x$ or $\sqrt{b+\sin 2x}$. It is so because we can manipulate $\sin 2x$ as $(\sin x+\cos x)^2 - 1$ or $1-(\sin x-\cos x)^2$ as per requirement.

(2) However if $\cos x \pm \sin x$ appears in denominator, then manipulate differently

Ex. $\int \frac{\cos x + \sin x}{2 - \sin 2x} dx$

Sol. $\cos x + \sin x = \frac{d}{dx}(\sin x - \cos x)$

So, write sin 2x as $1 - (\sin x - \cos x)^2$

and take sin x – cos x = t

$$\Rightarrow \int \frac{dt}{2 - (1 - t^2)} = \int \frac{dt}{1 + t^2}$$

$$= \tan^{-1}(t) + C$$

$$= \tan^{-1}(\sin x - \cos x) + C$$

Q. **Solve the following :**

(i) $\int \frac{dx}{\text{cosec } x + \cos x}$

Sol. $\int \frac{\sin x \, dx}{1 + \sin x \cos x}$

$$= \frac{1}{2}\underbrace{\int\left(\frac{\sin x + \cos x}{1 + \sin x \cos x}\right)dx}_{I_1} + \frac{1}{2}\underbrace{\int\left(\frac{\sin x - \cos x}{1 + \sin x \cos x}\right)dx}_{I_2}$$

In I_1, take $\sin x - \cos x = t$ and $\sin x \cos x = \frac{1 - t^2}{2}$

In I_2, take $\sin x + \cos x = y$ and $\sin x \cos x = \frac{y^2 - 1}{2}$

Now solve them separately and add to get answer.

Q. **(ii)** $\int \frac{\cos x dx}{\sqrt{8 - \sin 2x}}$

Sol. Let $I = I_1 + I_2$, where

$$I_1 = \frac{1}{2}\int \frac{(\cos x + \sin x)}{\sqrt{8 - \sin 2x}} dx \text{ and } I_2 = \frac{1}{2}\int \frac{(\cos x - \sin x)}{\sqrt{8 - \sin 2x}} dx$$

In I_1, $\sin x - \cos x = t$

$$\Rightarrow I_1 = \frac{1}{2}\int \frac{dt}{\sqrt{8 - (1 - t^2)}} = \frac{1}{2}\ell n\left|t + \sqrt{t^2 + 7}\right| + C_1$$

In I_2, $\sin x + \cos x = y$

$$\Rightarrow \quad I_2 = \frac{1}{2}\int \frac{dt}{\sqrt{8-\left(y^2-1\right)}} = \frac{1}{2}\sin^{-1}\left(\frac{y}{3}\right) + C_2$$

$I = I_1 + I_2$.

Integration of Irrational Algebraic Function

Type-1 :

(a) $\displaystyle\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(\beta-x)}} \quad (\beta > \alpha)$

Working Rule: Start with $x = \alpha\cos^2\theta + \beta\sin^2\theta$

(b) $\displaystyle\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}} \Big/ \int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

Working Rule: Start with $x = \alpha\sec^2\theta - \beta\tan^2\theta$

(c) $\displaystyle\int \sqrt{\frac{\beta-x}{x-\alpha}}dx$

Working Rule: Start with $x = \alpha\cos^2\theta + \beta\sin^2\theta$

Ex. $\displaystyle\int \sqrt{\frac{5-x}{x-2}}dx$

Sol. $\displaystyle\int \frac{5-x}{\sqrt{(5-x)(x-2)}}dx$

$x = 2\cos^2\theta + 5\sin^2\theta$

$\Rightarrow \quad dx = 3\sin 2\theta d\theta$

$$\Rightarrow \int \frac{\left(5-2\cos^2\theta-5\sin^2\theta\right)3\sin 2\theta\, d\theta}{\sqrt{\left(5-2\cos^2\theta-5\sin^2\theta\right)\left(2\cos^2\theta+5\sin^2\theta-2\right)}}$$

$$= \int \frac{3\cos^2\theta \times 3\sin 2\theta}{\sqrt{3\cos^2\theta \times 3\sin^2\theta}}d\theta$$

$$= \int \frac{3\cos^2\theta \times 2\sin\theta\cos\theta}{\sin\theta\cos\theta}d\theta = \int 6\cos^2\theta d\theta$$

$$= 3\int (1+\cos 2\theta)d\theta$$

$$= 3\left(\theta + \frac{\sin 2\theta}{2}\right) + C$$

Now $x = 2 + 3\sin^2\theta$

$\Rightarrow \sqrt{\frac{x-2}{3}} = \sin\theta$ and $\sqrt{\frac{5-x}{3}} = \cos\theta$

$\therefore \quad I = 3\sin^{-1}\sqrt{\frac{x-2}{3}} + \sqrt{(x-2)(5-x)} + C$

Type-2 :

$$\int \frac{dx}{(ax+b)\sqrt{px+q}}$$

Working Rule: $px + q = t^2$

Type-3 :

$$\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}; \text{e.g.} \int \frac{dx}{(x+1).\sqrt{1+x-x^2}}$$

Working Rule: Put $ax + b = \frac{1}{t}$

Type-4 :

$$\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}};$$

Working Rule: put $px + q = t^2$

e.g. $\int \frac{dx}{(x^2+5x+2)\sqrt{x-2}}$ this reduces to $\int \frac{dt}{t^4+9t^2+16}$

Type-5 :

$$\int \frac{dx}{(ax^2+bx+c)\sqrt{px^2+qx+r}}$$

Case-I :

When $(ax^2 + bx + c)$ breaks up into two linear factors, e.g.

$I = \int \frac{dx}{(x^2-x-2)\sqrt{x^2+x+1}}$ then

$$I = \int \left(\frac{A}{x-2} + \frac{B}{x+1}\right)\frac{1}{\sqrt{x^2+x+1}}dx = A\underbrace{\int \frac{dx}{(x-2)\sqrt{x^2+x+1}}}_{\text{put } x-2=\frac{1}{t}} + B\underbrace{\int \frac{dx}{(x+1)\sqrt{x^2+x+1}}}_{\text{put } x+1=\frac{1}{t}}$$

Case-II :

If $ax^2 + bx + c$ is a perfect square say $(lx+m)^2$

then put $lx+m=\frac{1}{t}$.

Case-III :

If $b = 0, q = 0$ e.g. $\int \frac{dx}{(ax^2+b)\sqrt{px^2+r}}$ then put

$x=\frac{1}{t}$ or the trigonometric substitution are also helpful.

e.g. $\int \frac{dx}{(x^2+4)\sqrt{4x^2+1}}$.

Ex. $\int \frac{dx}{(x+2)\sqrt{x+1}}$

Sol. $I=\int \frac{dx}{(x+2)\sqrt{x+1}}$

Put $x+1=t^2 \Rightarrow I=\int \frac{2t\,dt}{(t^2+1)t} = 2\tan^{-1}t + c = 2\tan^{-1}\left(\sqrt{x+1}\right)+c$

Ex. $\int \frac{dx}{(x^2+4)\sqrt{4x^2+1}}$

Sol. $I=\int \frac{dx}{(x^2+4)\sqrt{4x^2+1}}$

Put $x=\frac{1}{t}$ to get $I=\int \frac{-\frac{1}{t^2}dt}{\left(\frac{1}{t^2}+4\right)\sqrt{\frac{4}{t^2}+1}} = \int \frac{-tdt}{(1+4t^2)\sqrt{4+t^2}}$

Again put $4+t^2=z^2 \Rightarrow tdt=zdz \Rightarrow I=\int \frac{-zdz}{\{1+4(z^2-4)\}z}$

$$= \int \frac{-dz}{4z^2-15} = \frac{-1}{4}\int \frac{dz}{z^2-\frac{15}{4}} = -\frac{1}{4}\cdot\frac{1}{\sqrt{15}}\ln\left|\frac{2z-\sqrt{15}}{2z+\sqrt{15}}\right| + c$$

Now just put value of z and t.

Ex. $\int \frac{dx}{(x+1)\sqrt{1+x-x^2}}$

Sol. $I = \int \frac{dx}{(x+1)\sqrt{1+x-x^2}}$

Put $x+1 = \frac{1}{t} \Rightarrow I = \int \frac{-\frac{1}{t^2}dt}{\frac{1}{t}\sqrt{\frac{1}{t}-\left(\frac{1}{t}-1\right)^2}} = \int \frac{-dt}{\sqrt{3t-1-t^2}}$

$$= \int \frac{-dt}{\sqrt{\frac{5}{4}-\left(t-\frac{3}{2}\right)^2}} = -\sin^{-1}\left(\frac{2t-3}{\sqrt{5}}\right)+c = -\sin^{-1}\left(\frac{\frac{2}{x+1}-3}{\sqrt{5}}\right)+c$$

$$= \sin^{-1}\left(\frac{3x+1}{\sqrt{5}(x+1)}\right)+c$$

Q. **Solve the following :**

(i) $\int \frac{dx}{(2x+2)\sqrt{2x+1}}$

Sol. $2x+1 = y^2$

$\Rightarrow dx = y\,dy$

$\Rightarrow \int \frac{ydy}{(y^2+1)y}$

$= \tan^{-1} y + c$

$= \tan^{-1}\left(\sqrt{2x+1}\right) + C$

Q. **(ii)** $\int \frac{dx}{(x^2+2x+2)\sqrt{x+1}}$

Sol. $x+1 = y^2$

$\Rightarrow \int \frac{2ydy}{(y^4+1)y}$

$= \int \frac{1+y^2}{y^4+1}dy + \int \frac{1-y^2}{y^4+1}dy$

$$= \int \frac{1+\frac{1}{y^2}}{y^2+\frac{1}{y^2}}dy + \int \frac{\frac{1}{y^2}-1}{y^2+\frac{1}{y^2}}dy$$

$$= \int \frac{1+\frac{1}{y^2}}{\left(y-\frac{1}{y}\right)^2+2}dy - \int \frac{1-\frac{1}{y^2}}{\left(y+\frac{1}{y}\right)^2-2}dy$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{y-\frac{1}{y}}{\sqrt{2}}\right) - \frac{1}{2\sqrt{2}}\ell n\left|\frac{y+\frac{1}{y}-\sqrt{2}}{y+\frac{1}{y}+\sqrt{2}}\right| + C$$

Where $y = \sqrt{x+1}$

Q. **(iii)** $\int \frac{dx}{(x+1)\sqrt{x^2+2x+2}}$

Sol. $x+1 = \frac{1}{y}$

$$\int\left(\frac{-1}{y^2}\right)\frac{dy}{\left(\frac{1}{y}\right)\sqrt{\frac{1}{y^2}+1}}$$

$$= \int \frac{-dy}{\sqrt{1+y^2}} = -\ell n\left|y+\sqrt{1+y^2}\right| + C$$

Where $x+1 = \frac{1}{y}$

Q. **(iv)** $\int \frac{dx}{(x+1)(x+2)\sqrt{x^2+2x+2}}$

Sol. $$\int\left(\frac{1}{x+1}-\frac{1}{x+2}\right)\frac{dx}{\sqrt{x^2+2x+2}}$$

$$= \int \frac{dx}{(x+1)\sqrt{x^2+2x+2}} - \int \frac{dx}{(x+2)\sqrt{x^2+2x+2}}$$

Now it becomes just like previous question.

Q. (v) $\int \frac{dx}{(x^2-6x+9)\sqrt{x^2-6x+4}}$

Sol. $\int \frac{dx}{(x-3)^2\sqrt{(x-3)^2-5}}$

Let $x-3=\frac{1}{y}$

$$\Rightarrow \int\left(\frac{-1}{y^2}\right)\frac{dy}{\left(\frac{1}{y}\right)^2\sqrt{\left(\frac{1}{y}\right)^2-5}}$$

$$=\int\frac{-y\,dy}{\sqrt{1-5y^2}}$$

$$=\frac{1}{10}\int\frac{-10y\,dy}{\sqrt{1-5y^2}}$$

$$=\frac{1}{10}\times 2\times\sqrt{1-5y^2}+C$$

$$=\frac{\sqrt{1-5y^2}}{5}+C$$

$$=\frac{\sqrt{1-\frac{5}{(x-3)^2}}}{5}+C$$

Q. (vi) $\int \frac{dx}{(x^2+2)\sqrt{x^2+1}}$

Sol. **Method-I**

$x=\tan\theta$

$$\Rightarrow \int\frac{\sec^2\theta\,d\theta}{(\tan^2\theta+2)\sec\theta}$$

$$\Rightarrow \int\frac{\cos\theta\,d\theta}{\sin^2\theta+2\cos^2\theta}$$

$$=\int\frac{\cos\theta\,d\theta}{2-\sin^2\theta}$$

Let $\sin\theta = t$

$$\Rightarrow \quad \int \frac{dt}{2-t^2} = \frac{-1}{2\sqrt{2}} \ell n \left| \frac{t-\sqrt{2}}{t+\sqrt{2}} \right| + C$$

$$= \frac{-1}{2\sqrt{2}} \ell n \left| \frac{\sin\theta - \sqrt{2}}{\sin\theta + \sqrt{2}} \right| + C$$

Where $\tan\theta = x$

Q. **(vii)** $\int \frac{dx}{\left(x^2+4\right)\sqrt{4x^2+1}}$

Sol. $\int \frac{dx}{x^2\left(1+\frac{4}{x^2}\right)x\sqrt{4+\frac{1}{x^2}}}$

Let $4 + \frac{1}{x^2} = y^2$

$$\Rightarrow \quad \frac{-2}{x^3} dx = 2ydy$$

$$\Rightarrow \quad \int \frac{-y\,dy}{\left(1+4\left(y^2-4\right)\right)y}$$

$$= \int \frac{-dy}{4y^2-15}$$

$$= \frac{-1}{4} \int \frac{dy}{y^2 - \left(\frac{\sqrt{15}}{2}\right)^2}$$

$$= \frac{-1}{4\sqrt{15}} \ell n \left| \frac{2y-\sqrt{15}}{2y+\sqrt{15}} \right| + C$$

Where $y = \sqrt{4+\frac{1}{x^2}}$

N^{th} Integration

Ex. $\int (\tan x)^n \, dx$

Sol. Let $I_n = \int (\tan x)^n \, dx$

$$I_n = \int (\tan x)^{n-2} \left(\tan^2 x\right) dx$$

$$\Rightarrow \quad I_n = \int (\tan x)^{n-2} \sec^2 x \, dx - I_{n-2}$$

$$I_n = \frac{1}{n-1} (\tan x)^{n-1} - I_{n-2}$$

Ex. $\int (\cos x)^n \, dx$

Sol. $I_n = \int (\cos x)^{n-1} . \cos x dx$

$$I_n = (\cos x)^{n-1} \sin x + (n-1) \int (\cos x)^{n-2} \sin^2 x dx$$

$$\Rightarrow \quad I_n = (\cos x)^{n-1} \sin x + (n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow \quad n I_n = (\cos x)^{n-1} \sin x + (n-1) I_{n-2}$$

Q1 **If** $I_n = \int (\sin x)^n \, dx$ **, then** $5I_4 - 6I_6 = ?$

(A) $\sin x \cos^5 x + C$

(B) $\sin 2x \cos 2x$

(C) $\sin 2x \cos 4x$

(D) $\frac{\sin 2x}{8} \left(1 + \cos^2 2x - 2 \cos 2x\right)$

Sol. **(D)**

$$I_n = \int (\sin x)^{n-1} \sin x dx$$

$$\Rightarrow I_n = -\cos x (\sin x)^{n-1} + (n-1) \int (\sin x)^{n-2} \cos^2 x dx$$

$$\Rightarrow I_n = -\cos x (\sin x)^{n-1} + (n-1) I_{n-2} - (n-1) I_n + C$$

$$\Rightarrow n I_n - (n-1) I_{n-2} = -\cos x (\sin x)^{n-1} + C$$

Put $n = 6$

$$\Rightarrow 5I_4 - 6I_6 = \cos x (\sin x)^5 + C$$

$$= \frac{\sin 2x}{2}(\sin x)^4 + C$$

$$= \frac{\sin 2x}{8}(1 - \cos 2x)^2 + C$$

$$= \frac{\sin 2x}{8}(1 + \cos^2 2x - 2\cos 2x) + C$$

Q2 **Solve** $\int (\sec x)^n \, dx$

Sol. $I_n = \int (\sec x)^{n-2} \sec^2 x \, dx$

$$\Rightarrow I_n = (\sec x)^{n-2} \tan x - (n-2)\int (\sec x)^{n-3}(\sec x \tan x)\tan x \, dx$$

$$\Rightarrow I_n = (\sec x)^{n-2} \tan x - (n-2)\int (\sec x)^{n-2}(\sec^2 x - 1)\, dx$$

$$\Rightarrow (n-1)I_n = (\sec x)^{n-2} \tan x + (n-2)I_{n-2}$$